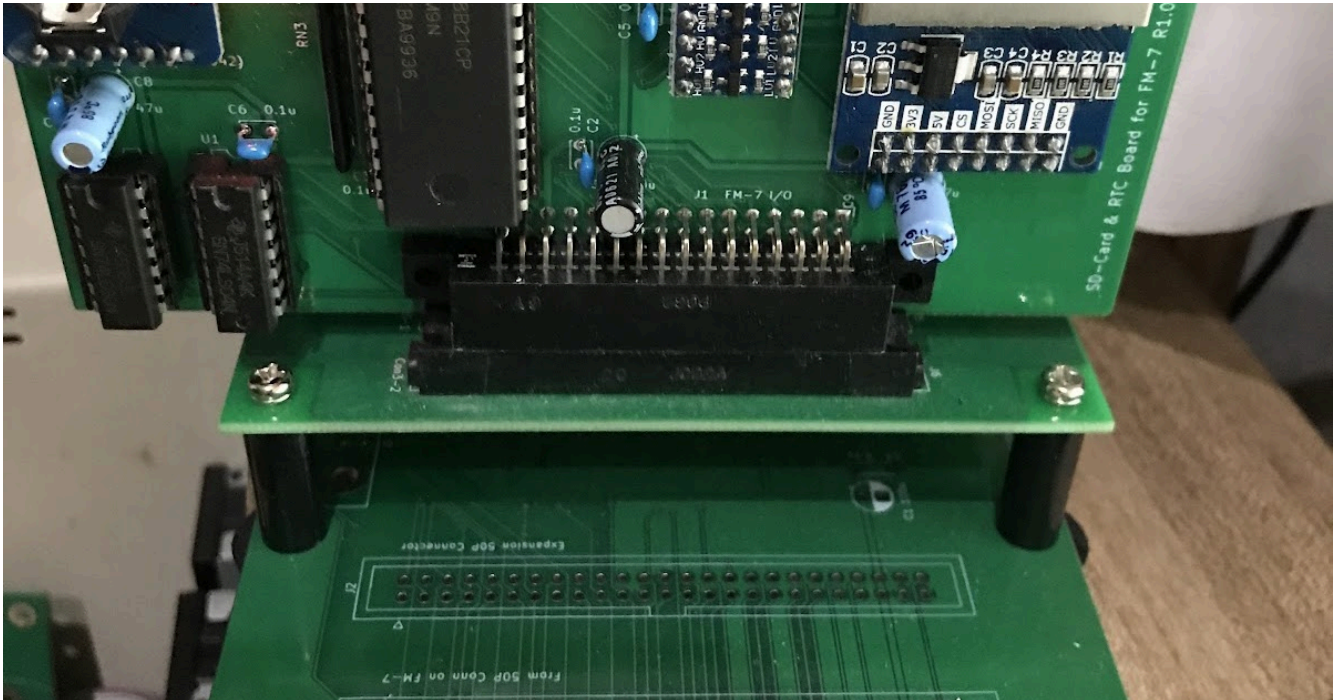


# An attempt to connect RTC and SD card to FM-7 (SD card version, Part 1)

 [flexonsbd.blogspot.com/2023/03/fm-7rtcandsd.html](https://flexonsbd.blogspot.com/2023/03/fm-7rtcandsd.html)

Thursday, March 16, 2023



## An attempt to connect an RTC and SD card to the FM-7 (SD card version, Part 1)

**I finally managed to read and write sectors to SD cards (including SDHC cards).**

It's still a work in progress, but I've managed to get it working to read and write the necessary sectors.

In addition to reading and writing any sector, it can read the MBR (Master Boot Record), BPB (Boot Parameter Block), FAT (File Allocation Table), DIR (Directory), and the first sector of a specified file. Of course, it works with SDHC cards as well as SD cards.

First, here's how it connects to the FM-7. In the previous report, it was mounted on the FM-7 unit itself, but this time it's mounted on the expansion board. The FDC card and FT245 card are lined up behind it, and they are connected to the unit's expansion connector with a 50-pin flat cable.

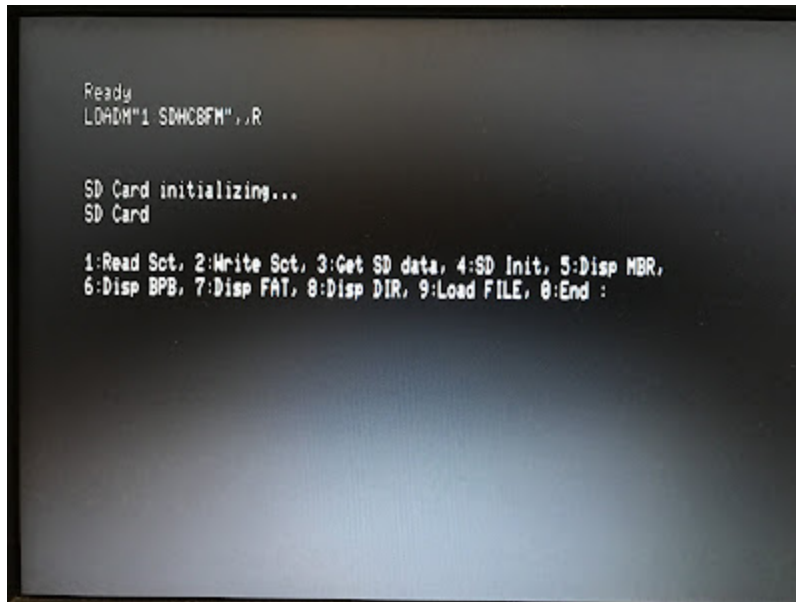


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Installation status

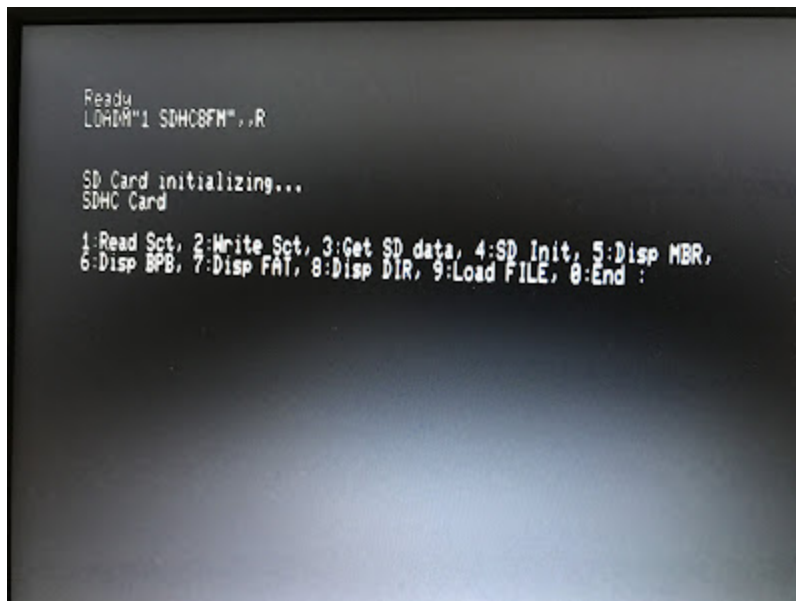
This is the boot screen when using an SD card.

A menu will appear after the SD card has been initialized and its type has been identified.



SD card boot screen

This is the boot screen when using an SDHC card.



SDHC card boot screen

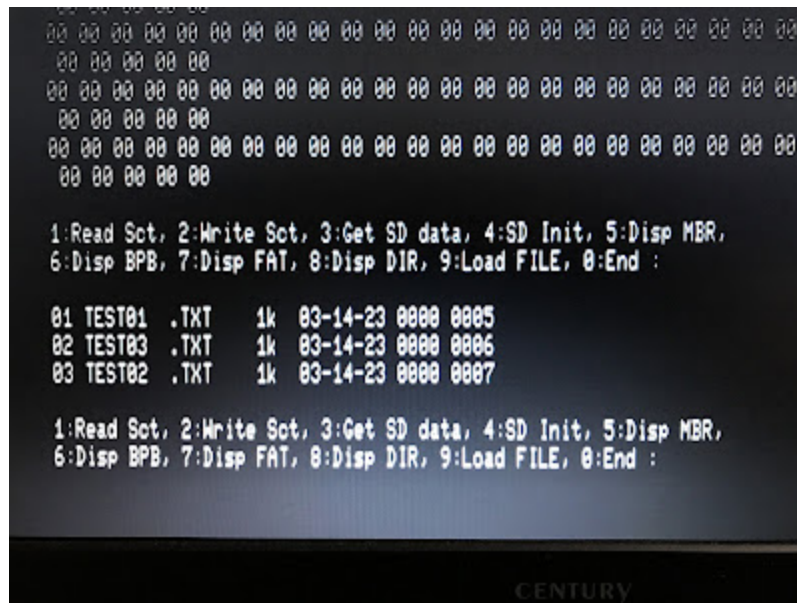
1. Read the specified sector
2. Write to the specified sector
- . 3. Acquire and display data from the SD card.
4. Initialize the SD card
5. Displaying the contents of the MBR (Master Boot Record)
6. Displaying the contents of the BPB (Boot Parameter Block)

7. Displaying the contents of the FAT (File Allocation Table)
8. Displaying the contents of the DIR (Directory)... (Only the first sector is displayed)
9. Display the file contents... (only the first sector)

As an example of menu execution, let's display the contents of the DIR directory.

The file name, size, creation date, cluster rank (top), and cluster rank (bottom) are displayed.

The leading number is used in step 9, "Displaying File Contents."



Displaying the directory on the SD card

This is menu item number 9.

Entering a file number will display the contents of that file. (Currently, this is a test program, so only the first sector is displayed.)

This screen is for an SDHC card, and the incomprehensible display on the third line is because lowercase letters are used in the filename on the fourth line. (The extension was .Lst)

Therefore, while FAT32 can be used, there is a limitation: lowercase letters cannot be used in filenames.

```
Reads
LOADM"1:SDHC8FM"...R

SD Card initializing...
SDHC Card

1:Read Sct, 2:Write Sct, 3:Get SD data, 4:SD Init, 5:Disp MBR,
6:Disp BPB, 7:Disp FAT, 8:Disp DIR, 9:Load FILE, 0:End :

01 TEST01 .TXT 1k 03-14-23 0000 0005
02 TEST02 .TXT 1k 03-14-23 0000 0007
03 ASDHC.7 ---k 03-19-06 0006 0000
04 SDHC7 .LST 19k 03-13-23 0000 0000
05 SDHC7 .TXT 1k 03-13-23 0000 0010
06 SDHC71 .LST 19k 03-13-23 0000 0011

Select File No :
```

---

Waiting for file number input.

Because the FM-7 has a limit of 80 characters per line, the 512 bytes of data in one sector do not fit neatly on a single screen, so there is no image of the display result. Instead, I will show the screen on FLEX9, which is used for development.

```
COM3000bps - Text Term V1
ファイルの 編集 設定 上一步-上一步 下一步-下一步 実行/停止

M8009 MONITOR Ver0.3
>
FLEX
8809 FLEX V3.01

Time : 2023/03/16 17:43:39

***1.SDHC.BIN
SD Card Initializing...
SDHC Card

1:Read Sct, 2:Write Sct, 3:Get SD data, 4:SD Init, 5:Disp MBR,
6:Disp BPB, 7:Disp FAT, 8:Disp DIR, 9:Load FILE, 0:End : 9

01 TEST01 .TXT 1k 08-14-23 0000 0005
02 TEST02 .TXT 1k 08-14-23 0000 0007
03 ASDNC.7 ---k 08-10-06 00:8 0000
04 SDHC7 .LST 18k 08-15-23 0000 0009
05 SDHC7 .TXT 1k 08-15-23 0000 0010
06 SDHC71 .LST 18k 08-15-23 0000 0011

Select File No : 5

2A 00 0A 2A 20 20 53 44 48 43 36 2E 54 58 54 00 0A 2A 00 0A 2A 20 20 53 44 20 52 85 61 04 2F 57
72 89 74 65 20 6F 6E 20 53 42 43 20 66 6F 72 20 46 4C 45 58 39 00 0A 2A 00 0A 2A 20 20 20 20 20 75
73 85 20 52 54 43 20 26 20 53 44 20 43 61 72 64 00 0A 2A 00 0A 2A 20 20 20 20 20 36 38 32 31 28 58
49 41 28 20 50 6F 72 74 20 41 20 3C 20 20 3E 20 52 54 43 20 28 49 32 43 20 40 6F 84 65 29 00 0A
2A 00 0A 2A 20 20 20 20 20 20 20 20 20 53 42 43 20 20 20 20 20 52 54 43 20 20 20 20 20 20 20 20 20
20 20 20 20 20 20 20 20 20 20 20 20 20 45 78 74 20 49 4F 00 0A 2A 20 20 20 20 20 43 41 32 20 20 32 29
20 20 20 3E 10 20 20 20 20 20 20 20 20 3E 20 20 48 56 34 20 4C 56 34 20 20 20 3E 20 20 49 4F 31
00 0A 2A 20 20 20 50 41 30 20 20 32 20 20 20 20 3E 20 20 53 43 4C 20 20 20 20 3E 20 20 48 56
31 20 4C 56 31 20 20 20 3E 20 20 49 4F 34 00 0A 2A 20 20 20 20 50 41 31 20 20 34 20 20 3C 20 20
20 20 20 20 20 20 20 20 20 20 20 20 48 56 32 20 4C 56 32 20 3C 20 20 20 49 4F 33 00 0A 2A 20
20 20 20 50 41 37 20 31 30 20 20 20 20 3E 20 20 53 44 41 20 20 20 20 3E 20 20 48 56 33 20 4C 56
53 20 20 20 3E 20 20 49 4F 32 00 0A 2A 20 20 20 20 47 4E 44 20 20 20 20 20 20 20 20 20 20 20
20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
20 20 20 33 58 33 00 0A 2A 00 0A 2A 00 0A 2A 20 20 20 36 38 32 31 28 50 49 41 29 20 50 6F 72
74 20 42 20 3C 20 2D 3E 20 53 44 20 28 53 50 49 20 40 6F 64 65 29 00 0A 2A 00 0A 2A 20 20 20 20
```

How it works on FLEX9

The above is a description of the loading function, which is one of the features that we have been able to implement so far.

Lowercase letters cannot be used in filenames, and of course folders cannot be used either, but since the goal is implementation on the FM-7, neither of these is a problem.

The problem is, even if you can read text and other data saved on an SD card on a Windows PC or similar device using the FM-7, how do you use that data within the FM-7's system?

The simplest method would be to load the data into memory, then read it from F-BASIC and save it to a file. This might suffice for small amounts of data, but...

Ideally, I'd like to be able to assign the SD card to drive 2 and use functions like FILES, LOAD, and SAVE, but achieving that seems quite difficult...

Also, the program size is likely to be quite large, so figuring out how to integrate it is another challenge.

It's still far from complete, but I think I'll use the subroutines I've created this time as a base to finish it, so I'll introduce the program I've created, even though it's roughly made.

```

1      0001      *
2      0002      * SDHC8FM.TXT
3      0003      *
4      0004      * SD Read/Write on SBC for FM-7
5      0005      *
6      0006      * use RTC & SD Card
7      0007      *
8      0008      * 6821(PIA) Port A ---- RTC (I2C Mode)
9      0009      *
10     0010      *          SBC          RTC          Ext IO
11     0011      * CA2  2  ---          ---  HV4 LV4 --- I01
12     0012      * PA0  3  ---  SCL  ---  HV1 LV1 --- I04
13     0013      * PA1  4  ---          ---  HV2 LV2 --- I03
14     0014      * PA7 10  ---  SDA  ---  HV3 LV3 --- I02
15     0015      * GND    --          --          -- GND
16     0016      *          --          --          -- 3V3
17     0017      *
18     0018      *
19     0019      * 6821(PIA) Port B ---- SD (SPI Mode)
20     0020      *
twenty 0021      *          SBC          SD
one    0022      * CB2 12  ---  HV4 LV4 --- SCK
twenty 0023      * PB0 13  ---  HV1 LV1 --- CS
two    0024      * PB1 14  ---  HV2 LV2 --- MISO
twenty 0025      * PB7 20  ---  HV3 LV3 --- MOSI
three  0026      * GND 31  --          -- GND
twenty 0027      * VCC 33  --          -- VCC
four   0028      *
twenty 0029      * cf. all data line use level converter
five   0030      *
26     0031      * PB output mode
27     0032      * CB2 1bit output
28     0033      *
29     0034      * READ/WRITE BUFFER $6E00 - 6FFF
30     0035      *
31     0036      * Add command          '2023.03.14
32     0037      * Change use Port B    '2023.03.07
33     0038      * Original version    '2022.06.11

```

```

34      0039      *      Copyright by Y.Yamamoto
35      0040      *
36      0041  6000      STTADR  EQU      $6000
37      0042  FBB8      INCHA   EQU      $FBB8
38      0043  FB8B      OUTCHA  EQU      $FB8B
39      0044
40      0045  6E00      RWBUF   EQU      STTADR+$E00
41      0046
42      0047  FD9E      PBDAT   EQU      $FD9E
43      0048  FD9F      PBCTL   EQU      PBDAT+1
44      0049
45      0050  6000      ORG      STTADR
46      0051
47      0052  6000  20  27  START  BRA      MAIN
48      0053  6002  01   VN      FCB      1
49      0054  6003  00   SCTN0   FCB      0      top adrs of sct
50      0055  6004  00   SCTN1   FCB      0
51      0056  6005  00   SCTN2   FCB      0
52      0057  6006  00   SCTN3   FCB      0
53      0058  6007  00   RESDAT  FCB      0      response data
54      0059  6008  00   SDHC    FCB      0      sd:0,sdhc:1
55      0060
56      0061  6009  0000      SCTBPB  FDB      00      sct no of
57      BPB(4bytes)
58      0062  600B  0000      FDB      00
59      0063  600D  0000      SCTFAT  FDB      00      sct no of
60      FAT#1(4bytes)
61      0064  600F  0000      FDB      00
62      0065  6011  0000      SCTDIR  FDB      00      sct no of
63      DIR(4bytes)
64      0066  6013  0000      FDB      00
65      0067  6015  0000      SCTCLT  FDB      00      sct no of top
66      CLUSTER(4bytes)
67      0068  6017  0000      FDB      00      if SD:#2, SDHC:#3
68      0069  6019  00   SCTPCL  FCB      0      sct per cluster
69      0070  601A  FF   WMDY    RMB      3
70      0071  601D  FF   WFSIZ   RMB      4
71      0072  6021  0000      CLSTH   FDB      00
72      0073  6023  0000      CLSTL   FDB      00

```

```

73      0074  6025  0000    WK1    FDB    00
74      0075  6027  0000    WK2    FDB    00
75      0076
76      0077  6029          MAIN    EQU    *
77      0078          *=====reset pia=====
78      0079          *portB:output
79      0080  6029  86  30          LDA    #$30
80      0081  602B  B7  FD9F          STA    PBCTL        select ddra,
81      cb2:0
82      0082  602E  86  FD          LDA    #$FD        pb0,7:out,pb1:in
83      0083  6030  B7  FD9E          STA    PBDAT        set output mode
84      0084  6033  86  34          LDA    #$34
85      0085  6035  B7  FD9F          STA    PBCTL        select
86      peripheral,cb2:0
87      0086
88      0087          *for pia exist check
89      0088  6038  86  55          LDA    #$55        test data
90      0089  603A  B7  FD9E          STA    PBDAT
91      0090  603D  B6  FD9E          LDA    PBDAT
92      0091  6040  84  FD          ANDA    #$FD        pb1=0
93      0092  6042  81  55          CMPA    #$55
94      0093  6044  27  07          BEQ     INIT
95      0094  6046  8E  68DF          LDX     #MERR
96      0095  6049  17  067B          LBSR    PSTRNG
97      0096  604C  39          RTS        return FM-7
98      0097
99      0098  604D          INIT    EQU    *
100     0099  604D  8D  5E          BSR     SDINIT
101     0100
102     0101          *****
103     0102          * MAIN MENU
104     0103          *****
105     0104  604F          MENU    EQU    *
106     0105  604F  8E  6817          LDX     #MSMENU1
107     0106  6052  17  0672          LBSR    PSTRNG
108     0107  6055  8E  6859          LDX     #MSMENU2
109     0108  6058  17  066C          LBSR    PSTRNG
110     0109
111     0110  605B  BD  FBB8          JSR     INCHA

```

|         |      |      |    |      |      |        |      |
|---------|------|------|----|------|------|--------|------|
| 112     | 0111 | 605E | 80 | 30   | SUBA | #\$30  |      |
| 113     | 0112 | 6060 | 26 | 04   | BNE  | MEN1   | 0?   |
| 114     | 0113 | 6062 | BD | 6612 | JSR  | CRLF   |      |
| 115     | 0114 | 6065 | 39 |      | RTS  | return | FM-7 |
| 116     | 0115 | 6066 | 4A | MEN1 | DECA |        |      |
| 117     | 0116 | 6067 | 26 | 05   | BNE  | MEN2   | 1?   |
| 118     | 0117 | 6069 | 17 | 0271 | LBSR | SDRD   |      |
| 119     | 0118 | 606C | 20 | E1   | BRA  | MENU   |      |
| 120     | 0119 | 606E | 4A | MEN2 | DECA |        |      |
| 121     | 0120 | 606F | 26 | 05   | BNE  | MEN3   | 2?   |
| 122     | 0121 | 6071 | 17 | 0416 | LBSR | SDWRT  |      |
| one two | 0122 | 6074 | 20 | D9   | BRA  | MENU   |      |
| three   | 0123 | 6076 | 4A | MEN3 | DECA |        |      |
| 124     | 0124 | 6077 | 26 | 05   | BNE  | MEN4   | 3?   |
| 125     | 0125 | 6079 | 17 | 01D3 | LBSR | GETSD  |      |
| 126     | 0126 | 607C | 20 | D1   | BRA  | MENU   |      |
| 127     | 0127 | 607E | 4A | MEN4 | DECA |        |      |
| 128     | 0128 | 607F | 26 | 04   | BNE  | MEN5   | 4?   |
| 129     | 0129 | 6081 | 8D | 2A   | BSR  | SDINIT |      |
| 130     | 0130 | 6083 | 20 | CA   | BRA  | MENU   |      |
| 131     | 0131 | 6085 | 4A | MEN5 | DECA | 5?     |      |
| 132     | 0132 | 6086 | 26 | 05   | BNE  | MEN6   |      |
| 133     | 0133 | 6088 | 17 | 0265 | LBSR | DSPMBR |      |
| 134     | 0134 | 608B | 20 | C2   | BRA  | MENU   |      |
| 135     | 0135 | 608D | 4A | MEN6 | DECA | 6?     |      |
| 136     | 0136 | 608E | 26 | 05   | BNE  | MEN7   |      |
| 137     | 0137 | 6090 | 17 | 02B7 | LBSR | DSPBPB |      |
| 138     | 0138 | 6093 | 20 | BA   | BRA  | MENU   |      |
| 139     | 0139 | 6095 | 4A | MEN7 | DECA | 7?     |      |
| 140     | 0140 | 6096 | 26 | 05   | BNE  | MEN8   |      |
| 141     | 0141 | 6098 | 17 | 02C1 | LBSR | DSPFAT |      |
| 142     | 0142 | 609B | 20 | B2   | BRA  | MENU   |      |
| 143     | 0143 | 609D | 4A | MEN8 | DECA | 8?     |      |
| 144     | 0144 | 609E | 26 | 05   | BNE  | MEN9   |      |
| 145     | 0145 | 60A0 | 17 | 02CB | LBSR | DSPDIR |      |
| 146     | 0146 | 60A3 | 20 | AA   | BRA  | MENU   |      |
| 147     | 0147 | 60A5 | 4A | MEN9 | DECA | 9?     |      |
| 148     | 0148 | 60A6 | 26 | A7   | BNE  | MENU   |      |
| 149     | 0149 | 60A8 | 17 | 0349 | LBSR | LDFILE |      |

```

150      0150  60AB  20  A2          BRA      MENU
151      0151          *
152      0152          *****
153      0153          * SD Initialize routine
154      0154          *****
155      0155  60AD          SDINIT  EQU      *
156      0156          *----- start -----
157      0157  60AD  8E  68C3          LDX      #MSINIT
158      0158  60B0  BD  66C7          JSR      PSTRNG
159      0159          * cs:1,di:1,sck-74ms
160      0160  60B3  86  FF          LDA      #$FF
161      mosi:b7=H,miso:b1=H(for PIA),cs:b0=H
162      0161  60B5  B7  FD9E          STA      PBDAT
163      0162  60B8  C6  50          LDB      #80      80times
164      0163  60BA  17  051A  ISTT    LBSR      SETCKL
165      0164  60BD  17  0520          LBSR      SETCKH
166      0165  60C0  5A          DECB
167      0166  60C1  26  F7          BNE      ISTT
168      0167          *
169      0168          *----- cmd 0 -----
170      0169  60C3          SCMD0    EQU      *
171      0170          * LDX #MCMD0
172      0171          * LBSR PSTRNG
173      0172
174      0173  60C3  17  0511          LBSR      SETCKL
175      0174  60C6  C6  06          LDB      #6
176      0175  60C8  8E  67B3          LDX      #CMD0
177      0176  60CB  A6  80  SC01    LDA      ,X+
178      0177  60CD  17  0496          LBSR      SNDDAT
179      0178  60D0  5A          DECB
180      0179  60D1  26  F8          BNE      SC01
181      0180          *get response
182      0181  60D3  5F          CLRB
183      0182  60D4          GR00
184      0183  60D4  5A          DECB
185      0184  60D5  1027  01FD          LBEQ      SDERR
186      0185  60D9  17  04A6          LBSR      GETDAT
187      0186  60DC  B6  6007          LDA      RESDAT
188      0187          * LBSR DSPBIN disp response

```

```

189      0188  60DF  81  01          CMPA    #$01
190      0189  60E1  26  F1          BNE     GR00
191      0190          * JSR CRLF
192      0191          *
193      0192          *----- cmd 8 -----
194      0193  60E3          SCMD8    EQU     *
195      0194          * LDX #MCMD8
196      0195          * LBSR PSTRNG
197      0196
198      0197  60E3  17  04F1          LBSR    SETCKL
199      0198  60E6  86  FF          LDA     #$FF      dummy cmd
200      0199  60E8  17  047B          LBSR    SNDDAT
201      0200  60EB  C6  06          LDB     #6
202      0201  60ED  8E  67CB          LDX     #CMD8
203      0202  60F0  A6  80  SC80      LDA     ,X+
204      0203  60F2  17  0471          LBSR    SNDDAT
205      0204  60F5  5A          DECB
206      0205  60F6  26  F8          BNE     SC80
207      0206          *get response
208      0207  60F8          GR80
209      0208  60F8  17  0487          LBSR    GETDAT
210      0209  60FB  B6  6007          LDA     RESDAT
211      0210
212      0211  60FE  17  0481          LBSR    GETDAT
213      0212  6101  B6  6007          LDA     RESDAT
214      0213          * LBSR DSPBIN disp response
215      0214  6104  81  01          CMPA    #$01
216      0215  6106  26  F0          BNE     GR80
217      0216  6108  C6  04          LDB     #4
218      0217  610A  17  0475  GR81    LBSR    GETDAT
219      0218  610D  B6  6007          LDA     RESDAT
220      0219          * LBSR DSPBIN disp response
221      0220  6110  5A          DECB
222      0221  6111  26  F7          BNE     GR81
223      0222          * JSR CRLF
224      0223          *
225      0224          *----- cmd 58 -----
226      0225  6113          SCMD58    EQU     *
227      0226          * LDX #MCMD58

```

|     |      |      |    |            |                             |        |           |
|-----|------|------|----|------------|-----------------------------|--------|-----------|
| 228 | 0227 |      |    |            | * LBSR PSTRNG               |        |           |
| 229 | 0228 |      |    |            |                             |        |           |
| 230 | 0229 | 6113 | 17 | 04C1       | LBSR                        | SETCKL |           |
| 231 | 0230 | 6116 | 86 | FF         | LDA                         | #\$FF  | dummy cmd |
| 232 | 0231 | 6118 | 17 | 044B       | LBSR                        | SNDDAT |           |
| 233 | 0232 | 611B | C6 | 06         | LDB                         | #6     |           |
| 234 | 0233 | 611D | 8E | 67D1       | LDX                         | #CMD58 |           |
| 235 | 0234 | 6120 | A6 | 80 SC581   | LDA                         | ,X+    |           |
| 236 | 0235 | 6122 | 17 | 0441       | LBSR                        | SNDDAT |           |
| 237 | 0236 | 6125 | 5A |            | DECB                        |        |           |
| 238 | 0237 | 6126 | 26 | F8         | BNE                         | SC581  |           |
| 239 | 0238 |      |    |            | *get response               |        |           |
| 240 | 0239 | 6128 |    | GR580      |                             |        |           |
| 241 | 0240 | 6128 | 17 | 0457       | LBSR                        | GETDAT |           |
| 242 | 0241 | 612B | B6 | 6007       | LDA                         | RESDAT |           |
| 243 | 0242 |      |    |            | * LBSR DSPBIN disp response |        |           |
| 244 | 0243 | 612E | 81 | 01         | CMPA                        | #\$01  |           |
| 245 | 0244 | 6130 | 26 | F6         | BNE                         | GR580  |           |
| 246 | 0245 | 6132 | C6 | 04 GR581   | LDB                         | #4     |           |
| 247 | 0246 | 6134 | 17 | 044B GR582 | LBSR                        | GETDAT |           |
| 248 | 0247 | 6137 | B6 | 6007       | LDA                         | RESDAT |           |
| 249 | 0248 |      |    |            | * LBSR DSPBIN disp response |        |           |
| 250 | 0249 | 613A | 5A |            | DECB                        |        |           |
| 251 | 0250 | 613B | 26 | F7         | BNE                         | GR582  |           |
| 252 | 0251 | 613D |    | RPT41 EQU  | *                           |        |           |
| 253 | 0252 |      |    |            | *----- cmd 55 -----         |        |           |
| 254 | 0253 | 613D |    | SCMD55 EQU | *                           |        |           |
| 255 | 0254 |      |    |            | * LDX #MCMD55               |        |           |
| 256 | 0255 |      |    |            | * LBSR PSTRNG               |        |           |
| 257 | 0256 |      |    |            |                             |        |           |
| 258 | 0257 | 613D | 17 | 0497       | LBSR                        | SETCKL |           |
| 259 | 0258 | 6140 | 86 | FF         | LDA                         | #\$FF  | dummy cmd |
| 260 | 0259 | 6142 | 17 | 0421       | LBSR                        | SNDDAT |           |
| 261 | 0260 | 6145 | C6 | 06         | LDB                         | #6     |           |
| 262 | 0261 | 6147 | 8E | 67D7       | LDX                         | #CMD55 |           |
| 263 | 0262 | 614A | A6 | 80 SC551   | LDA                         | ,X+    |           |
| 264 | 0263 | 614C | 17 | 0417       | LBSR                        | SNDDAT |           |
| 265 | 0264 | 614F | 5A |            | DECB                        |        |           |
| 266 | 0265 | 6150 | 26 | F8         | BNE                         | SC551  |           |

```

267      0266      *get response
268      0267      * CLR B
269      0268 6152      GR550
270      0269      * DEC B
271      0270      * BEQ SDERR
272      0271 6152 17 042D      LBSR      GETDAT
273      0272 6155 B6 6007      LDA      RESDAT
274      0273      * LBSR DSPBIN disp response
275      0274 6158 81 01      CMPA      #$01
276      0275 615A 26 F6      BNE      GR550
277      0276
278      0277      * JSR CRLF
279      0278      *----- acmd 41 -----
280      0279 615C      ASCMD41 EQU      *
281      0280      * LDX #MCMD41
282      0281      * LBSR PSTRNG
283      0282
284      0283 615C 17 0478      LBSR      SETCKL
285      0284 615F 86 FF      LDA      #$FF      dummy cmd
286      0285 6161 17 0402      LBSR      SNDDAT
287      0286 6164 C6 06      LDB      #6
288      0287 6166 8E 67DD      LDX      #ACMD41
289      0288 6169 A6 80 ASC411 LDA      ,X+
290      0289 616B 17 03F8      LBSR      SNDDAT
291      0290 616E 5A      DECB
292      0291 616F 26 F8      BNE      ASC411
293      0292      *get response
294      0293 6171      GR410
295      0294 6171 17 040E      LBSR      GETDAT
296      0295 6174 B6 6007      LDA      RESDAT
297      0296      * LBSR DSPBIN disp response
298      0297 6177 17 0408      LBSR      GETDAT
299      0298 617A B6 6007      LDA      RESDAT
300      0299      * LBSR DSPBIN disp response
301      0300 617D 81 00      CMPA      #$00
302      0301 617F 26 BC      BNE      RPT41
303      0302      * JSR CRLF
304      0303
305      0304      *----- second cmd 58 -----

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306      0305  6181      SCMD258 EQU      *
307      0306      * LDX #MCMD58
308      0307      * LBSR PSTRNG
309      0308
310      0309  6181  17  0453      LBSR      SETCKL
311      0310  6184  86  FF      LDA      #$FF      dummy cmd
312      0311  6186  17  03DD      LBSR      SNDDAT
313      0312  6189  C6  06      LDB      #6
314      0313  618B  8E  67D1      LDX      #CMD58
315      0314  618E  A6  80  SC2581  LDA      ,X+
316      0315  6190  17  03D3      LBSR      SNDDAT
317      0316  6193  5A      DECB
318      0317  6194  26  F8      BNE      SC2581
319      0318      *get response
320      0319  6196      GR2580
321      0320  6196  17  03E9      LBSR      GETDAT
322      0321  6199  B6  6007      LDA      RESDAT
323      0322      * LBSR DSPBIN disp response
324      0323  619C  81  00      CMPA     #$00
325      0324  619E  26  F6      BNE      GR2580
326      0325  61A0  17  03DF      LBSR      GETDAT
327      0326  61A3  B6  6007      LDA      RESDAT
328      0327      * LBSR DSPBIN disp response
329      0328  61A6  48      LSLA
330      0329  61A7  48      LSLA
331      0330  61A8  24  03      BCC      GR2581
332      0331  61AA  7C  6008      INC      SDHC      if sdhc then 1
333      0332  61AD  C6  03  GR2581  LDB      #3
334      0333  61AF  17  03D0  GR2582  LBSR      GETDAT
335      0334  61B2  B6  6007      LDA      RESDAT
336      0335      * LBSR DSPBIN disp response
337      0336  61B5  5A      DECB
338      0337  61B6  26  F7      BNE      GR2582
339      0338
340      0339  61B8  8E  68A9      LDX      #MSSD0
341      0340  61BB  B6  6008      LDA      SDHC
342      0341  61BE  27  03      BEQ      SD1
343      0342  61C0  8E  68B5      LDX      #MSSD1
344      0343  61C3  17  0501  SD1      LBSR      PSTRNG

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345      0344
346      0345      *-----
347      0346      *set sct no of BPB
348      0347  61C6  17  0145      LBSR      GETMBR
349      0348  61C9  8E  6FCA      LDX      #RWBUF+446+8+4
350      0349  61CC  108E  6009      LDY      #SCTBPB
351      0350  61D0  C6  04      LDB      #4
352      0351  61D2  A6  82  ILOP1  LDA      ,-X
353      0352  61D4  A7  A0      STA      ,Y+
354      0353  61D6  5A      DECB
355      0354  61D7  26  F9      BNE      ILOP1
356      0355
357      0356      *set sct no of FAT#1
358      0357  61D9  17  0155      LBSR      GETBPB
359      0358  61DC  8E  6009      LDX      #SCTBPB
360      0359  61DF  108E  600D      LDY      #SCTFAT
361      0360  61E3  C6  04      LDB      #4
362      0361  61E5  A6  80  ILOP2  LDA      ,X+
363      0362  61E7  A7  A0      STA      ,Y+
364      0363  61E9  5A      DECB
365      0364  61EA  26  F9      BNE      ILOP2
366      0365
367      0366  61EC  8E  600D      LDX      #SCTFAT
368      0367  61EF  108E  6E0E      LDY      #RWBUF+$0E      for SD & SDHC
369      0368  61F3  17  04A0      LBSR      ADD42
370      0369
371      0370      *set sct no of DIR
372      0371  61F6  8E  600D      LDX      #SCTFAT
373      0372  61F9  108E  6011      LDY      #SCTDIR
374      0373  61FD  C6  04      LDB      #4
375      0374  61FF  A6  80  ILOP3  LDA      ,X+
376      0375  6201  A7  A0      STA      ,Y+
377      0376  6203  5A      DECB
378      0377  6204  26  F9      BNE      ILOP3
379      0378
380      0379  6206  8E  6E16      LDX      #RWBUF+$16      for SD
381      0380  6209  7D  6008      TST      SDHC
382      0381  620C  27  03      BEQ      SKP1
383      0382  620E  C6  0E      LDB      #$24-$16

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384      0383  6210  3A          ABX          for SDHC
385      0384  6211  1F  12  SKP1    TFR      X,Y
386      0385  6213  8E  6011          LDX      #SCTDIR
387      0386  6216  17  047D          LBSR      ADD42
388      0387  6219  17  047A          LBSR      ADD42          skip for FAT#2
389      0388
390      0389          *set sct per cluster
391      0390  621C  8E  6E0D          LDX      #RWBUF+$0D
392      0391  621F  A6  84          LDA      ,X
393      0392  6221  B7  6019          STA      SCTPCL
394      0393
395      0394          *set sct no of CLUSTER#2
396      0395  6224  8E  6011          LDX      #SCTDIR
397      0396  6227  108E  6015          LDY      #SCTCLT
398      0397  622B  C6  04          LDB      #4
399      0398  622D  A6  80  IL0P4    LDA      ,X+
400      0399  622F  A7  A0          STA      ,Y+
401      0400  6231  5A          DECB
402      0401  6232  26  F9          BNE      IL0P4
403      0402
404      0403  6234  108E  6027          LDY      #WK1+2
405      0404  6238  86  00          LDA      #$00
406      0405  623A  A7  A2          STA      ,-Y
407      0406  623C  86  20          LDA      #$20          for SD
408      0407  623E  7D  6008          TST      SDHC
409      0408  6241  27  03          BEQ      SKP2
410      0409  6243  B6  6019          LDA      SCTPCL          for SDHC
411      0410  6246  A7  A2  SKP2    STA      ,-Y
412      0411  6248  8E  6015          LDX      #SCTCLT
413      0412  624B  17  0448          LBSR      ADD42
414      0413          *-----
415      0414  624E  39          RTS
416      0415
417      0416          *****
418      0417          * GET SD DATA ROUTINE
419      0418          * use CMD9
420      0419          *****
421      0420  624F          GETSD    EQU      *
422      0421  624F  BD  6612          JSR      CRLF

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|     |      |      |    |                                         |      |           |                 |
|-----|------|------|----|-----------------------------------------|------|-----------|-----------------|
| 423 | 0422 | 6252 | 17 | 0382                                    | LBSR | SETCKL    |                 |
| 424 | 0423 | 6255 | C6 | 06                                      | LDB  | #6        |                 |
| 425 | 0424 | 6257 | 8E | 67E3                                    | LDX  | #CMD9     |                 |
| 426 | 0425 | 625A | A6 | 80 SC91                                 | LDA  | ,X+       |                 |
| 427 | 0426 | 625C | 17 | 0307                                    | LBSR | SNDDAT    |                 |
| 428 | 0427 | 625F | 5A |                                         | DECB |           |                 |
| 429 | 0428 | 6260 | 26 | F8                                      | BNE  | SC91      |                 |
| 430 | 0429 |      |    | * JSR CRLF                              |      |           |                 |
| 431 | 0430 |      |    | *get response                           |      |           |                 |
| 432 | 0431 | 6262 |    | GR90                                    |      |           |                 |
| 433 | 0432 | 6262 | 17 | 031D                                    | LBSR | GETDAT    |                 |
| 434 | 0433 | 6265 | B6 | 6007                                    | LDA  | RESDAT    |                 |
| 435 | 0434 |      |    | * LBSR DSPBIN disp response             |      |           |                 |
| 436 | 0435 | 6268 | 81 | 00                                      | CMPA | #\$00     |                 |
| 437 | 0436 | 626A | 26 | F6                                      | BNE  | GR90      |                 |
| 438 | 0437 |      |    |                                         |      |           |                 |
| 439 | 0438 |      |    | * JSR CRLF                              |      |           |                 |
| 440 | 0439 | 626C | BD | 6612                                    | JSR  | CRLF      |                 |
| 441 | 0440 | 626F | C6 | 14                                      | LDB  | #16+1+2+1 | fe,crcH,crcL,ff |
| 442 | 0441 | 6271 | 17 | 030E GR91                               | LBSR | GETDAT    |                 |
| 443 | 0442 | 6274 | B6 | 6007                                    | LDA  | RESDAT    |                 |
| 444 | 0443 | 6277 | 17 | 036F                                    | LBSR | DSPBIN    | disp response   |
| 445 | 0444 | 627A | 5A |                                         | DECB |           |                 |
| 446 | 0445 | 627B | 26 | F4                                      | BNE  | GR91      |                 |
| 447 | 0446 | 627D | BD | 6612                                    | JSR  | CRLF      |                 |
| 448 | 0447 |      |    |                                         |      |           |                 |
| 449 | 0448 |      |    | *disp offset of MBR,FAT#1,DIR,CLUSTER#2 |      |           |                 |
| 450 | 0449 | 6280 | 8E | 6905                                    | LDX  | #MSOFS    |                 |
| 451 | 0450 | 6283 | BD | 66C7                                    | JSR  | PSTRNG    |                 |
| 452 | 0451 | 6286 | 8E | 6917                                    | LDX  | #MSBPB    |                 |
| 453 | 0452 | 6289 | BD | 66C7                                    | JSR  | PSTRNG    |                 |
| 454 | 0453 | 628C | 8E | 6009                                    | LDX  | #SCTBPB   |                 |
| 455 | 0454 | 628F | C6 | 04                                      | LDB  | #4        |                 |
| 456 | 0455 | 6291 | A6 | 80 SCL0P1                               | LDA  | ,X+       |                 |
| 457 | 0456 | 6293 | 17 | 0353                                    | LBSR | DSPBIN    |                 |
| 458 | 0457 | 6296 | 5A |                                         | DECB |           |                 |
| 459 | 0458 | 6297 | 26 | F8                                      | BNE  | SCL0P1    |                 |
| 460 | 0459 |      |    |                                         |      |           |                 |
| 461 | 0460 | 6299 | 8E | 6920                                    | LDX  | #MSFAT    |                 |

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462      0461  629C  BD  66C7          JSR      PSTRNG
463      0462  629F  8E  600D          LDX      #SCTFAT
464      0463  62A2  C6  04          LDB      #4
465      0464  62A4  A6  80  SCL0P2  LDA      ,X+
466      0465  62A6  17  0340          LBSR     DSPBIN
467      0466  62A9  5A          DECB
468      0467  62AA  26  F8          BNE      SCL0P2
469      0468
470      0469  62AC  8E  6929          LDX      #MSDIR
471      0470  62AF  BD  66C7          JSR      PSTRNG
472      0471  62B2  8E  6011          LDX      #SCTDIR
473      0472  62B5  C6  04          LDB      #4
474      0473  62B7  A6  80  SCL0P3  LDA      ,X+
475      0474  62B9  17  032D          LBSR     DSPBIN
476      0475  62BC  5A          DECB
477      0476  62BD  26  F8          BNE      SCL0P3
478      0477
479      0478  62BF  8E  6932          LDX      #MSCL2
480      0479  62C2  BD  66C7          JSR      PSTRNG
481      0480  62C5  8E  6015          LDX      #SCTCLT
482      0481  62C8  C6  04          LDB      #4
483      0482  62CA  A6  80  SCL0P4  LDA      ,X+
484      0483  62CC  17  031A          LBSR     DSPBIN
485      0484  62CF  5A          DECB
486      0485  62D0  26  F8          BNE      SCL0P4
487      0486  62D2  BD  6612          JSR      CRLF
488      0487  62D5  39          RTS
489      0488
490      0489          *
491      0490  62D6          SDERR  EQU      *
492      0491  62D6  8E  68DF          LDX      #MERR
493      0492  62D9  17  03EB          LBSR     PSTRNG
494      0493  62DC          OKEND
495      0494  62DC  39          RTS          return FM-7
496      0495          *
497      0496          *****
498      0497          * READ ROUTINE
499      0498          * use CMD17
500      0499          * set sctno to $6005,6006

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501      0500      * read data set to $6800-69FF
502      0501      *****
503      0502      62DD      SDRD      EQU      *
504      0503      62DD      BD      6612      JSR      CRLF
505      0504      62E0      17      02B9      LBSR      INSCT
506      0505      62E3      17      0227      LBSR      RD1SCT      read 1 sct
507      0506
508      0507      *disp data
509      0508      62E6      BD      6612      JSR      CRLF
510      0509      62E9      17      025C      LBSR      DSPBUF
511      0510      62EC      BD      6612      JSR      CRLF
512      0511      62EF      39      RTS
513      0512
514      0513      *****
515      0514      * Read MBR & Disp parameter of BPB
516      0515      *****
517      0516      62F0      DSPMBR      EQU      *
518      0517      62F0      BD      6612      JSR      CRLF
519      0518      62F3      8D      19      BSR      GETMBR
520      0519
521      0520      *disp BPB
522      0521      62F5      BD      6612      JSR      CRLF
523      0522      62F8      8E      6FBE      LDX      #RWBUF+446
524      0523      62FB      C6      10      GET1      LDB      #16
525      0524      62FD      A6      80      GET2      LDA      ,X+
526      0525      62FF      17      02E7      LBSR      DSPBIN      disp data
527      0526      6302      5A      DECB
528      0527      6303      26      F8      BNE      GET2
529      0528      6305      BD      6612      JSR      CRLF
530      0529      6308      8C      6FFE      CMPX      #RWBUF+510      446+16*4
531      0530      630B      25      EE      BLO      GET1
532      0531      630D      39      RTS
533      0532
534      0533      *-----
535      0534      *read MBR
536      0535      630E      GETMBR      EQU      *
537      0536      630E      8E      6003      LDX      #SCTN0
538      0537      6311      C6      04      LDB      #4
539      0538      6313      4F      CLRA

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540      0539  6314  A7  80  GPL0P  STA      ,X+
541      0540  6316  5A              DECB
542      0541  6317  26  FB              BNE      GPL0P
543      0542  6319  17  0288          LBSR      SETSCT
544      0543
545      0544  631C  17  01EE          LBSR      RD1SCT      read 1 sct
546      0545  631F  39              RTS
547      0546
548      0547      *-----
549      0548      *set sector no to SCTN0
550      0549      *from X,X+1,X+2,X+3 to SCTN0-SCTN3
551      0550  6320      SETSN0  EQU      *
552      0551  6320  108E  6003          LDY      #SCTN0
553      0552  6324  C6  04              LDB      #4
554      0553  6326  A6  80  SEL0P  LDA      ,X+
555      0554  6328  A7  A0              STA      ,Y+
556      0555  632A  5A              DECB
557      0556  632B  26  F9              BNE      SEL0P
558      0557  632D  17  0274          LBSR      SETSCT
559      0558  6330  39              RTS
560      0559
561      0560      *-----
562      0561      *read BPB
563      0562  6331      GETBPB  EQU      *
564      0563  6331  8D  DB              BSR      GETMBR
565      0564      *set sct no of BPB
566      0565  6333  8E  6FCA          LDX      #RWBUF+446+8+4
567      0566  6336  108E  6003          LDY      #SCTN0
568      0567  633A  C6  04              LDB      #4
569      0568  633C  A6  82  GPL0P2  LDA      , -X
570      0569  633E  A7  A0              STA      ,Y+
571      0570  6340  5A              DECB
572      0571  6341  26  F9              BNE      GPL0P2
573      0572  6343  17  025E          LBSR      SETSCT
574      0573
575      0574  6346  17  01C4          LBSR      RD1SCT      read 1 sct
576      0575  6349  39              RTS
577      0576
578      0577      *****

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579      0578      * Disp BPB(Bios Parameter Block)
580      0579      *****
581      0580      634A      DSPBPB  EQU      *
582      0581      634A      8E      6009      LDX      #SCTBPB
583      0582      634D      8D      D1      BSR      SETSN0
584      0583      634F      17      01BB      LBSR      RD1SCT      read 1 sct
585      0584
586      0585      *disp BPB
587      0586      6352      BD      6612      JSR      CRLF
588      0587      6355      17      01F0      LBSR      DSPBUF
589      0588      6358      BD      6612      JSR      CRLF
590      0589      635B      39      RTS
591      0590
592      0591      *****
593      0592      * Disp FAT#1
594      0593      *****
595      0594      635C      DSPFAT  EQU      *
596      0595      635C      8E      600D      LDX      #SCTFAT
597      0596      635F      8D      BF      BSR      SETSN0
598      0597      6361      17      01A9      LBSR      RD1SCT      read 1 sct
599      0598
600      0599      *disp FAT
601      0600      6364      BD      6612      JSR      CRLF
602      0601      6367      17      01DE      LBSR      DSPBUF
603      0602      636A      BD      6612      JSR      CRLF
604      0603      636D      39      RTS
605      0604
606      0605      *****
607      0606      * Disp Directory
608      0607      *****
609      0608      636E      DSPDIR  EQU      *
610      0609      636E      8E      6011      LDX      #SCTDIR
611      0610      6371      8D      AD      BSR      SETSN0
612      0611      6373      17      0197      LBSR      RD1SCT      read 1 sct
613      0612
614      0613      **disp DIR
615      0614      * JSR CRLF
616      0615      * LBSR DSPBUF
617      0616      * JSR CRLF

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|     |      |      |       |            |           |      |           |               |
|-----|------|------|-------|------------|-----------|------|-----------|---------------|
| 618 | 0617 |      | *     |            |           |      |           |               |
| 619 | 0618 |      | *disp | FILE       | name      |      |           |               |
| 620 | 0619 | 6376 | BD    | 6612       |           | JSR  | CRLF      |               |
| 621 | 0620 | 6379 | BD    | 6612       |           | JSR  | CRLF      |               |
| 622 | 0621 | 637C | 8E    | 6E60       |           | LDX  | #RWBUF+96 | skip top data |
| 623 | 0622 | 637F | 7F    | 6027       |           | CLR  | WK2       | counter       |
| 624 | 0623 | 6382 | A6    | 84         | DSPFL1    | LDA  | ,X        |               |
| 625 | 0624 | 6384 | 27    | 6D         |           | BEQ  | DSPEND    |               |
| 626 | 0625 | 6386 | 81    | E5         |           | CMPA | #\$E5     | delete mark?  |
| 627 | 0626 | 6388 | 26    | 05         |           | BNE  | DSPFL2    |               |
| 628 | 0627 | 638A | 30    | 88         | 20        | LEAX | 32,X      |               |
| 629 | 0628 | 638D | 20    | F3         |           | BRA  | DSPFL1    |               |
| 630 | 0629 |      |       |            |           |      |           |               |
| 631 | 0630 | 638F | 7C    | 6027       | DSPFL2    | INC  | WK2       |               |
| 632 | 0631 | 6392 | B6    | 6027       |           | LDA  | WK2       |               |
| 633 | 0632 | 6395 | 17    | 0251       |           | LBSR | DSPBIN    | disp counter  |
| 634 | 0633 | 6398 | C6    | 08         |           | LDB  | #8        |               |
| 635 | 0634 | 639A | A6    | 80         | DSPFL3    | LDA  | ,X+       |               |
| 636 | 0635 | 639C | BD    | FBBB       |           | JSR  | OUTCHA    |               |
| 637 | 0636 | 639F | 5A    |            |           | DECB |           |               |
| 638 | 0637 | 63A0 | 26    | F8         |           | BNE  | DSPFL3    |               |
| 639 | 0638 | 63A2 | 86    | 2E         |           | LDA  | #'.       |               |
| 640 | 0639 | 63A4 | BD    | FBBB       |           | JSR  | OUTCHA    |               |
| 641 | 0640 | 63A7 | C6    | 03         |           | LDB  | #3        |               |
| 642 | 0641 | 63A9 | A6    | 80         | DSPFL4    | LDA  | ,X+       |               |
| 643 | 0642 | 63AB | BD    | FBBB       |           | JSR  | OUTCHA    |               |
| 644 | 0643 | 63AE | 5A    |            |           | DECB |           |               |
| 645 | 0644 | 63AF | 26    | F8         |           | BNE  | DSPFL4    |               |
| 646 | 0645 | 63B1 | 86    | 20         |           | LDA  | #'        |               |
| 647 | 0646 | 63B3 | BD    | FBBB       |           | JSR  | OUTCHA    |               |
| 648 | 0647 | 63B6 | BD    | FBBB       |           | JSR  | OUTCHA    |               |
| 649 | 0648 |      |       |            |           |      |           |               |
| 650 | 0649 |      | *get  | FILE       | time data |      |           |               |
| 651 | 0650 | 63B9 | 30    | 07         |           | LEAX | 7,X       |               |
| 652 | 0651 | 63BB | EC    | 81         |           | LDD  | ,X++      |               |
| 653 | 0652 | 63BD | FD    | 6025       |           | STD  | WK1       |               |
| 654 | 0653 |      |       |            |           |      |           |               |
| 655 | 0654 |      | *get  | CLUSTER(H) | data      |      |           |               |
| 656 | 0655 | 63C0 | EC    | 81         |           | LDD  | ,X++      |               |

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657      0656  63C2  1E  89          EXG      A,B
658      0657  63C4  FD  6021          STD      CLSTH
659      0658  63C7  30  04          LEAX      4,X
660      0659  63C9  EC  81          LDD        ,X++
661      0660  63CB  1E  89          EXG      A,B
662      0661  63CD  FD  6023          STD      CLSTL
663      0662
664      0663          *get FILE size data
665      0664  63D0  108E  6021          LDY      #WFSIZ+4
666      0665  63D4  C6  04          LDB        #4
667      0666  63D6  A6  80  DSPC1  LDA        ,X+
668      0667  63D8  A7  A2          STA        , -Y
669      0668  63DA  5A          DECB
670      0669  63DB  26  F9          BNE        DSPC1
671      0670
672      0671          *disp FILE size
673      0672  63DD  86  20          LDA        #'
674      0673  63DF  BD  FBBB          JSR        OUTCHA
675      0674  63E2  17  0353          LBSR      DSPSIZ      disp FILE size
676      0675
677      0676
678      0677          *disp FILE time
679      0678  63E5  FC  6025          LDD        WK1
680      0679  63E8  17  02EC          LBSR      DSPYMD      disp MM-DD-YY
681      0680
682      0681          *disp CLUSTER(H,L)
683      0682  63EB  17  03A0          LBSR      DSPCLS
684      0683
685      0684  63EE  BD  6612          JSR        CRLF
686      0685  63F1  20  8F          BRA        DSPFL1
687      0686          *
688      0687  63F3  39          DSPEND  RTS
689      0688
690      0689          *****
691      0690          * Load FILE
692      0691          *****
693      0692  63F4          LDFILE  EQU      *
694      0693  63F4  17  FF77          LBSR      DSPDIR
695      0694  63F7  8E  6942          LDX        #MSINFL

```

|     |      |      |      |       |       |      |           |                           |
|-----|------|------|------|-------|-------|------|-----------|---------------------------|
| 696 | 0695 | 63FA | BD   | 66C7  |       | JSR  | PSTRNG    |                           |
| 697 | 0696 | 63FD | BD   | FBB8  |       | JSR  | INCHA     | file no                   |
| 698 | 0697 | 6400 | 80   | 30    |       | SUBA | #\$30     | to binary                 |
| 699 | 0698 |      |      |       |       |      |           |                           |
| 700 | 0699 | 6402 | 8E   | 6E60  |       | LDX  | #RWBUF+96 | skip top data             |
| 701 | 0700 | 6405 | 1F   | 89    |       | TFR  | A,B       |                           |
| 702 | 0701 |      |      |       |       |      |           |                           |
| 703 | 0702 | 6407 | A6   | 84    | LDL0P | LDA  | ,X        |                           |
| 704 | 0703 | 6409 | 81   | E5    |       | CMPA | #\$E5     | delete mark?              |
| 705 | 0704 | 640B | 26   | 05    |       | BNE  | LD1       |                           |
| 706 | 0705 | 640D | 30   | 88 20 |       | LEAX | 32,X      | skip                      |
| 707 | 0706 | 6410 | 20   | F5    |       | BRA  | LDL0P     |                           |
| 708 | 0707 | 6412 | 5A   | LD1   | DECB  |      |           |                           |
| 709 | 0708 | 6413 | 27   | 05    |       | BEQ  | LD2       |                           |
| 710 | 0709 | 6415 | 30   | 88 20 |       | LEAX | 32,X      |                           |
| 711 | 0710 | 6418 | 20   | ED    |       | BRA  | LDL0P     |                           |
| 712 | 0711 |      |      |       |       |      |           | *hit file, get cluster no |
| 713 | 0712 | 641A | EC   | 88 14 | LD2   | LDD  | \$14,X    |                           |
| 714 | 0713 | 641D | 1E   | 89    |       | EXG  | A,B       |                           |
| 715 | 0714 | 641F | FD   | 6021  |       | STD  | CLSTH     |                           |
| 716 | 0715 | 6422 | EC   | 88 1A |       | LDD  | \$1A,X    |                           |
| 717 | 0716 | 6425 | 1E   | 89    |       | EXG  | A,B       |                           |
| 718 | 0717 | 6427 | FD   | 6023  |       | STD  | CLSTL     |                           |
| 719 | 0718 |      |      |       |       |      |           | *get sct no               |
| 720 | 0719 | 642A | C0   | 02    |       | SUBB | #2        | sub top cluser no         |
| 721 | 0720 | 642C | 7D   | 6008  |       | TST  | SDHC      | SD:0,SDHC:1               |
| 722 | 0721 | 642F | 27   | 01    |       | BEQ  | LD3       |                           |
| 723 | 0722 | 6431 | 5A   |       | DECB  |      |           |                           |
| 724 | 0723 |      |      |       |       |      |           |                           |
| 725 | 0724 | 6432 | 34   | 04    | LD3   | PSHS | B         |                           |
| 726 | 0725 | 6434 | 8E   | 6015  |       | LDX  | #SCTCLT   |                           |
| 727 | 0726 | 6437 | 108E | 6025  |       | LDY  | #WK1      |                           |
| 728 | 0727 | 643B | C6   | 04    |       | LDB  | #4        |                           |
| 729 | 0728 | 643D | A6   | 80    | LD4   | LDA  | ,X+       |                           |
| 730 | 0729 | 643F | A7   | A0    |       | STA  | ,Y+       |                           |
| 731 | 0730 | 6441 | 5A   |       | DECB  |      |           |                           |
| 732 | 0731 | 6442 | 26   | F9    |       | BNE  | LD4       |                           |
| 733 | 0732 | 6444 | 35   | 04    |       | PULS | B         |                           |
| 734 | 0733 |      |      |       |       |      |           |                           |

|     |      |      |      |      |                          |        |            |
|-----|------|------|------|------|--------------------------|--------|------------|
| 735 | 0734 | 6446 | 8E   | 6025 | LDX                      | #WK1   |            |
| 736 | 0735 | 6449 | A6   | 03   | LD5                      | LDA    | 3,X        |
| 737 | 0736 | 644B | 8B   | 40   |                          | ADDA   | #\$40      |
| 738 | 0737 | 644D | A7   | 03   |                          | STA    | 3,X        |
| 739 | 0738 | 644F | 24   | 06   |                          | BCC    | LD51       |
| 740 | 0739 | 6451 | A6   | 02   |                          | LDA    | 2,X        |
| 741 | 0740 | 6453 | 8B   | 01   |                          | ADDA   | #1         |
| 742 | 0741 | 6455 | A7   | 02   |                          | STA    | 2,X        |
| 743 | 0742 | 6457 | 24   | 06   | LD51                     | BCC    | LD52       |
| 744 | 0743 | 6459 | A6   | 01   |                          | LDA    | 1,X        |
| 745 | 0744 | 645B | 8B   | 01   |                          | ADDA   | #1         |
| 746 | 0745 | 645D | A7   | 01   |                          | STA    | 1,X        |
| 747 | 0746 | 645F | 24   | 06   | LD52                     | BCC    | LD53       |
| 748 | 0747 | 6461 | A6   | 84   |                          | LDA    | ,X         |
| 749 | 0748 | 6463 | 8B   | 01   |                          | ADDA   | #1         |
| 750 | 0749 | 6465 | A7   | 84   |                          | STA    | ,X         |
| 751 | 0750 | 6467 | 5A   | LD53 | DECB                     |        |            |
| 752 | 0751 | 6468 | 26   | DF   | BNE                      | LD5    |            |
| 753 | 0752 |      |      |      |                          |        |            |
| 754 | 0753 | 646A | 8E   | 6025 | LDX                      | #WK1   |            |
| 755 | 0754 | 646D | 108E | 6003 | LDY                      | #SCTN0 |            |
| 756 | 0755 | 6471 | C6   | 04   | LDB                      | #4     |            |
| 757 | 0756 | 6473 | A6   | 80   | LD6                      | LDA    | ,X+        |
| 758 | 0757 | 6475 | A7   | A0   |                          | STA    | ,Y+        |
| 759 | 0758 | 6477 | 5A   |      | DECB                     |        |            |
| 760 | 0759 | 6478 | 26   | F9   | BNE                      | LD6    |            |
| 761 | 0760 | 647A | 17   | 0127 | LBSR                     | SETSCT |            |
| 762 | 0761 | 647D | 17   | 008D | LBSR                     | RD1SCT | read 1 sct |
| 763 | 0762 |      |      |      |                          |        |            |
| 764 | 0763 |      |      |      | * LDX #SCTCLT            |        |            |
| 765 | 0764 |      |      |      | * LBSR SETSNO            |        |            |
| 766 | 0765 |      |      |      | * LBSR RD1SCT read 1 sct |        |            |
| 767 | 0766 |      |      |      | *                        |        |            |
| 768 | 0767 |      |      |      | **disp CLUSTER#2         |        |            |
| 769 | 0768 | 6480 | BD   | 6612 | JSR                      | CRLF   |            |
| 770 | 0769 | 6483 | 17   | 00C2 | LBSR                     | DSPBUF |            |
| 771 | 0770 | 6486 | BD   | 6612 | JSR                      | CRLF   |            |
| 772 | 0771 | 6489 | 39   |      | RTS                      |        |            |
| 773 | 0772 |      |      |      |                          |        |            |

```

774      0773      *****
775      0774      * WRITE ROUTINE
776      0775      * use CMD24
777      0776      * set sctno to $6005,6006
778      0777      * write data from $6A00-6BFF
779      0778      *****
780      0779      648A      SDWRT      EQU      *
781      0780      648A      8E      689D      LDX      #MSWROK
782      0781      648D      17      0237      LBSR      PSTRNG
783      0782      6490      BD      FBB8      JSR      INCHA
784      0783      6493      84      DF      ANDA      #$DF
785      0784      6495      81      59      CMPA      #'Y
786      0785      6497      27      01      BEQ      SDWR1
787      0786      6499      39      RTS
788      0787      649A      BD      6612      SDWR1      JSR      CRLF
789      0788      649D      17      00FC      LBSR      INSCT      *input SCT No
790      0789      *copy sct no to cmd param
791      0790      64A0      8E      6003      LDX      #SCTN0
792      0791      64A3      108E      67C6      LDY      #CMD24+1
793      0792      64A7      C6      04      LDB      #4
794      0793      64A9      A6      80      CPY2      LDA      ,X+
795      0794      64AB      A7      A0      STA      ,Y+
796      0795      64AD      5A      DECB
797      0796      64AE      26      F9      BNE      CPY2
798      0797      *send cmd
799      0798      64B0      17      0124      LBSR      SETCKL
800      0799      64B3      C6      06      LDB      #6
801      0800      64B5      8E      67C5      LDX      #CMD24
802      0801      64B8      A6      80      SDW1      LDA      ,X+
803      0802      64BA      17      012C      LBSR      DSPBIN      disp response
804      0803      64BD      17      00A6      LBSR      SNDDAT
805      0804      64C0      5A      DECB
806      0805      64C1      26      F5      BNE      SDW1
807      0806      *get cmd response
808      0807      64C3      SDW2
809      0808      64C3      17      00BC      LBSR      GETDAT
810      0809      64C6      B6      6007      LDA      RESDAT
811      0810      64C9      17      011D      LBSR      DSPBIN      disp response
812      0811      64CC      81      00      CMPA      #$00

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|     |      |      |      |                    |        |        |                |
|-----|------|------|------|--------------------|--------|--------|----------------|
| 813 | 0812 | 64CE | 26   | F3                 | BNE    | SDW2   |                |
| 814 | 0813 |      |      | *1byte skip        |        |        |                |
| 815 | 0814 | 64D0 | 86   | FF                 | LDA    | #\$FF  |                |
| 816 | 0815 | 64D2 | 17   | 0091               | LBSR   | SNDDAT | 1byte skip     |
| 817 | 0816 | 64D5 | BD   | 6612               | JSR    | CRLF   |                |
| 818 | 0817 |      |      |                    |        |        |                |
| 819 | 0818 |      |      | *send data packet  |        |        |                |
| 820 | 0819 | 64D8 | 8E   | 6E00               | LDX    | #RWBUF | --temp buf     |
| 821 | 0820 | 64DB | 108E | 0200               | LDY    | #512   |                |
| 822 | 0821 |      |      | *send data token   |        |        |                |
| 823 | 0822 | 64DF | 86   | FE                 | LDA    | #\$FE  |                |
| 824 | 0823 | 64E1 | 17   | 0082               | LBSR   | SNDDAT | data token     |
| 825 | 0824 | 64E4 | A6   | 80                 | SNDDT1 | LDA    | ,X+            |
| 826 | 0825 | 64E6 | 8D   | 7E                 | BSR    | SNDDAT |                |
| 827 | 0826 | 64E8 | 31   | 3F                 | LEAY   | -1,Y   |                |
| 828 | 0827 | 64EA | 26   | F8                 | BNE    | SNDDT1 |                |
| 829 | 0828 |      |      |                    |        |        |                |
| 830 | 0829 | 64EC | 86   | FF                 | LDA    | #\$FF  |                |
| 831 | 0830 | 64EE | 8D   | 76                 | BSR    | SNDDAT | crcH           |
| 832 | 0831 | 64F0 | 8D   | 74                 | BSR    | SNDDAT | crcL           |
| 833 | 0832 |      |      | *get data response |        |        |                |
| 834 | 0833 | 64F2 |      | CKRS               |        |        |                |
| 835 | 0834 | 64F2 | 17   | 008D               | LBSR   | GETDAT |                |
| 836 | 0835 | 64F5 | B6   | 6007               | LDA    | RESDAT |                |
| 837 | 0836 | 64F8 | 17   | 00EE               | LBSR   | DSPBIN | disp response  |
| 838 | 0837 | 64FB | 81   | E5                 | CMPA   | #\$E5  | data response? |
| 839 | 0838 | 64FD | 26   | F3                 | BNE    | CKRS   |                |
| 840 | 0839 |      |      | *wait until busy   |        |        |                |
| 841 | 0840 | 64FF |      | BUSY               |        |        |                |
| 842 | 0841 | 64FF | 17   | 0080               | LBSR   | GETDAT |                |
| 843 | 0842 | 6502 | B6   | 6007               | LDA    | RESDAT |                |
| 844 | 0843 | 6505 | 17   | 00E1               | LBSR   | DSPBIN | disp response  |
| 845 | 0844 | 6508 | 81   | FF                 | CMPA   | #\$FF  |                |
| 846 | 0845 | 650A | 26   | F3                 | BNE    | BUSY   |                |
| 847 | 0846 | 650C | 39   |                    | RTS    |        |                |
| 848 | 0847 |      |      |                    |        |        |                |
| 849 | 0848 |      |      | *-----             |        |        |                |
| 850 | 0849 |      |      | *read 1 sct        |        |        |                |
| 851 | 0850 |      |      | *-----             |        |        |                |

```

852      0851  650D      RD1SCT EQU      *
853      0852          *copy sct no to cmd param
854      0853  650D  8E  6003          LDX      #SCTN0
855      0854  6510  108E  67C0          LDY      #CMD17+1
856      0855  6514  C6  04          LDB      #4
857      0856  6516  A6  80  CPY1      LDA      ,X+
858      0857  6518  A7  A0          STA      ,Y+
859      0858  651A  5A          DECB
860      0859  651B  26  F9          BNE      CPY1
861      0860          *send cmd
862      0861  651D  17  00B7          LBSR      SETCKL
863      0862  6520  C6  06          LDB      #6
864      0863  6522  8E  67BF          LDX      #CMD17
865      0864  6525  A6  80  RD1      LDA      ,X+
866      0865  6527  8D  3D          BSR      SNDDAT
867      0866  6529  5A          DECB
868      0867  652A  26  F9          BNE      RD1
869      0868          *get response
870      0869  652C      RD2
871      0870  652C  8D  54          BSR      GETDAT
872      0871  652E  B6  6007          LDA      RESDAT
873      0872  6531  81  FE          CMPA      #$FE      token?
874      0873  6533  26  F7          BNE      RD2
875      0874
876      0875          *get data
877      0876          * JSR CRLF
878      0877  6535  8E  6E00          LDX      #RWBUF      --temp buf
879      0878  6538  108E  0200          LDY      #512
880      0879  653C      RD3
881      0880  653C  8D  44          BSR      GETDAT
882      0881  653E  B6  6007          LDA      RESDAT
883      0882  6541  A7  80          STA      ,X+      save buf
884      0883  6543  31  3F          LEAY      -1,Y
885      0884  6545  26  F5          BNE      RD3
886      0885  6547  39          RTS
887      0886
888      0887          *-----
889      0888          *disp data
890      0889  6548      DSPBUF EQU      *

```

```

891      0890  6548  BD  6612          JSR      CRLF
892      0891  654B  8E  6E00          LDX      #RWBUF      --temp buf
893      0892  654E  108E  0200          LDY      #512
894      0893  6552  C6  20  DSPB1  LDB      #32
895      0894  6554          DSPB2
896      0895  6554  A6  80          LDA      ,X+      save buf
897      0896  6556  17  0090          LBSR      DSPBIN      disp data
898      0897  6559  31  3F          LEAY      -1,Y
899      0898  655B  27  08          BEQ      DSPB3
900      0899  655D  5A          DECB
901      0900  655E  26  F4          BNE      DSPB2
902      0901  6560  BD  6612          JSR      CRLF
903      0902  6563  20  ED          BRA      DSPB1
904      0903  6565  39          DSPB3  RTS
905      0904
906      0905          *-----
907      0906          *send data
908      0907  6566          SNDDAT  EQU      *
909      0908  6566  34  06          PSHS      A,B
910      0909  6568  7F  6007          CLR      RESDAT
911      0910  656B  C6  08          LDB      #8
912      0911  656D  34  02  SNDD1  PSHS      A
913      0912  656F  84  FC          ANDA      #$FC
914      0913  6571  8A  02          ORA      #$02      miso:b1=H(for
915      PIA),cs:b0=L
916      0914  6573  B7  FD9E          STA      PBDAT      send data
917      0915  6576  8D  5F          BSR      SETCKL
918      0916  6578  35  02          PULS      A
919      0917  657A  48          LSLA
920      0918  657B  8D  63          BSR      SETCKH
921      0919  657D  5A          DECB
922      0920  657E  26  ED          BNE      SNDD1
923      0921  6580  35  86          PULS      A,B,PC
924      0922
925      0923          *get data
926      0924  6582          GETDAT  EQU      *
927      0925  6582  34  06          PSHS      A,B
928      0926  6584  C6  08          LDB      #8
929      0927  6586  8D  4F  GETD1  BSR      SETCKL

```

|     |      |      |               |                       |      |        |                   |
|-----|------|------|---------------|-----------------------|------|--------|-------------------|
| 930 | 0928 | 6588 | 12            |                       | NOP  |        |                   |
| 931 | 0929 | 6589 | 12            |                       | NOP  |        |                   |
| 932 | 0930 | 658A | 12            |                       | NOP  |        |                   |
| 933 | 0931 | 658B | 8D            | 53                    | BSR  | SETCKH |                   |
| 934 | 0932 | 658D | B6            | FD9E                  | LDA  | PBDAT  | get response      |
| 935 | 0933 | 6590 | 44            |                       | LSRA |        |                   |
| 936 | 0934 | 6591 | 44            |                       | LSRA | b1--C  |                   |
| 937 | 0935 | 6592 | 79            | 6007                  | ROL  | RESDAT | C--b0             |
| 938 | 0936 | 6595 | 5A            |                       | DECB |        |                   |
| 939 | 0937 | 6596 | 26            | EE                    | BNE  | GETD1  |                   |
| 940 | 0938 | 6598 | 8D            | 3D                    | BSR  | SETCKL |                   |
| 941 | 0939 | 659A | 35            | 86                    | PULS | A,B,PC |                   |
| 942 | 0940 |      | *             |                       |      |        |                   |
| 943 | 0941 |      | *-----        |                       |      |        |                   |
| 944 | 0942 |      | *input SCT No |                       |      |        |                   |
| 945 | 0943 | 659C |               | INSCT EQU *           |      |        |                   |
| 946 | 0944 | 659C | 8E            | 6895                  | LDX  | #MSSCT |                   |
| 947 | 0945 | 659F | 17            | 0125                  | LBSR | PSTRNG |                   |
| 948 | 0946 | 65A2 | 8D            | 79                    | BSR  | GET8CH | get 8chr, convert |
| 949 | 4bin |      |               |                       |      |        |                   |
| 950 | 0947 |      |               |                       |      |        |                   |
| 951 | 0948 | 65A4 |               | SETSCT EQU *          |      |        |                   |
| 952 | 0949 | 65A4 | 7D            | 6008                  | TST  | SDHC   |                   |
| 953 | 0950 | 65A7 | 26            | 1D                    | BNE  | CALCHC |                   |
| 954 | 0951 |      |               | *SD card              |      |        |                   |
| 955 | 0952 | 65A9 | C6            | 03                    | LDB  | #3     |                   |
| 956 | 0953 | 65AB | 8E            | 6004                  | LDX  | #SCTN1 |                   |
| 957 | 0954 | 65AE | 108E          | 6003                  | LDY  | #SCTN0 |                   |
| 958 | 0955 | 65B2 | A6            | 80 SFT1               | LDA  | ,X+    |                   |
| 959 | 0956 | 65B4 | A7            | A0                    | STA  | ,Y+    |                   |
| 960 | 0957 | 65B6 | 5A            |                       | DECB |        |                   |
| 961 | 0958 | 65B7 | 26            | F9                    | BNE  | SFT1   |                   |
| 962 | 0959 | 65B9 | 7F            | 6006                  | CLR  | SCTN3  |                   |
| 963 | 0960 |      |               | *calc top adrs of sct |      |        |                   |
| 964 | 0961 | 65BC |               | CALCSD EQU *          |      |        |                   |
| 965 | 0962 | 65BC | 78            | 6006                  | LSL  | SCTN3  |                   |
| 966 | 0963 | 65BF | 79            | 6005                  | ROL  | SCTN2  |                   |
| 967 | 0964 | 65C2 | 79            | 6004                  | ROL  | SCTN1  |                   |
| 968 | 0965 | 65C5 | 39            |                       | RTS  |        |                   |

|      |      |      |           |                         |      |          |       |  |
|------|------|------|-----------|-------------------------|------|----------|-------|--|
| 969  | 0966 |      |           |                         |      |          |       |  |
| 970  | 0967 |      |           | *SDHC card              |      |          |       |  |
| 971  | 0968 |      |           | *calc top adrs of sct   |      |          |       |  |
| 972  | 0969 | 65C6 |           | CALCHC EQU *            |      |          |       |  |
| 973  | 0970 | 65C6 | 8E 6003   |                         | LDX  | #SCTN0   |       |  |
| 974  | 0971 | 65C9 | 108E 67C0 |                         | LDY  | #CMD17+1 |       |  |
| 975  | 0972 | 65CD | C6 04     |                         | LDB  | #4       |       |  |
| 976  | 0973 | 65CF | A6 80     | CALCH1                  | LDA  | ,X+      |       |  |
| 977  | 0974 | 65D1 | A7 A0     |                         | STA  | ,Y+      |       |  |
| 978  | 0975 | 65D3 | 5A        |                         | DECB |          |       |  |
| 979  | 0976 | 65D4 | 26 F9     |                         | BNE  | CALCH1   |       |  |
| 980  | 0977 | 65D6 | 39        |                         | RTS  |          |       |  |
| 981  | 0978 |      |           |                         |      |          |       |  |
| 982  | 0979 |      |           | *                       |      |          |       |  |
| 983  | 0980 |      |           | *set sck:L              |      |          |       |  |
| 984  | 0981 | 65D7 |           | SETCKL EQU *            |      |          |       |  |
| 985  | 0982 | 65D7 | 34 02     |                         | PSHS | A        |       |  |
| 986  | 0983 | 65D9 | 86 34     |                         | LDA  | #\$34    | ca2=L |  |
| 987  | 0984 | 65DB | B7 FD9F   |                         | STA  | PBCTL    |       |  |
| 988  | 0985 | 65DE | 35 82     |                         | PULS | A,PC     |       |  |
| 989  | 0986 |      |           |                         |      |          |       |  |
| 990  | 0987 |      |           | *set sck:H              |      |          |       |  |
| 991  | 0988 | 65E0 |           | SETCKH EQU *            |      |          |       |  |
| 992  | 0989 | 65E0 | 34 02     |                         | PSHS | A        |       |  |
| 993  | 0990 | 65E2 | 86 3C     |                         | LDA  | #\$3C    | ca2=H |  |
| 994  | 0991 | 65E4 | B7 FD9F   |                         | STA  | PBCTL    |       |  |
| 995  | 0992 | 65E7 | 35 82     |                         | PULS | A,PC     |       |  |
| 996  | 0993 |      |           |                         |      |          |       |  |
| 997  | 0994 |      |           | *binary(A)--ascii(A)(B) |      |          |       |  |
| 998  | 0995 | 65E9 |           | DSPBIN EQU *            |      |          |       |  |
| 999  | 0996 | 65E9 | 34 06     |                         | PSHS | A,B      |       |  |
| 1000 | 0997 | 65EB | 1F 89     |                         | TFR  | A,B      |       |  |
| 1001 | 0998 | 65ED | 44        |                         | LSRA |          |       |  |
| 1002 | 0999 | 65EE | 44        |                         | LSRA |          |       |  |
| 1003 | 1000 | 65EF | 44        |                         | LSRA |          |       |  |
| 1004 | 1001 | 65F0 | 44        |                         | LSRA |          |       |  |
| 1005 | 1002 | 65F1 | 8A 30     |                         | ORA  | #\$30    |       |  |
| 1006 | 1003 | 65F3 | C4 0F     |                         | ANDB | #\$0F    |       |  |
| 1007 | 1004 | 65F5 | CA 30     |                         | ORB  | #\$30    |       |  |

|      |      |      |            |        |      |      |         |
|------|------|------|------------|--------|------|------|---------|
| 1008 | 1005 | 65F7 | 81         | 39     |      | CMPA | #\$39   |
| 1009 | 1006 | 65F9 | 23         | 02     |      | BLS  | BI21    |
| 1010 | 1007 | 65FB | 8B         | 07     |      | ADDA | #\$07   |
| 1011 | 1008 | 65FD | C1         | 39     | BI21 | CMPB | #\$39   |
| 1012 | 1009 | 65FF | 23         | 02     |      | BLS  | BI22    |
| 1013 | 1010 | 6601 | CB         | 07     |      | ADDB | #\$07   |
| 1014 | 1011 | 6603 | BD         | FBBB   | BI22 | JSR  | OUTCHA  |
| 1015 | 1012 | 6606 | 1F         | 98     |      | TFR  | B,A     |
| 1016 | 1013 | 6608 | BD         | FBBB   |      | JSR  | OUTCHA  |
| 1017 | 1014 | 660B | 86         | 20     |      | LDA  | #\$20   |
| 1018 | 1015 | 660D | BD         | FBBB   |      | JSR  | OUTCHA  |
| 1019 | 1016 | 6610 | 35         | 86     |      | PULS | A,B,PC  |
| 1020 | 1017 |      |            |        |      |      |         |
| 1021 | 1018 |      | *crlf      |        |      |      |         |
| 1022 | 1019 | 6612 |            | CRLF   | EQU  | *    |         |
| 1023 | 1020 | 6612 | 86         | 0D     |      | LDA  | #\$0D   |
| 1024 | 1021 | 6614 | BD         | FBBB   |      | JSR  | OUTCHA  |
| 1025 | 1022 | 6617 | 86         | 0A     |      | LDA  | #\$0A   |
| 1026 | 1023 | 6619 | BD         | FBBB   |      | JSR  | OUTCHA  |
| 1027 | 1024 | 661C | 39         |        |      | RTS  |         |
| 1028 | 1025 |      |            |        |      |      |         |
| 1029 | 1026 |      | *          |        |      |      |         |
| 1030 | 1027 | 661D |            | GET8CH | EQU  | *    |         |
| 1031 | 1028 |      | *input chr |        |      |      |         |
| 1032 | 1029 | 661D | 8E         | 6956   |      | LDX  | #INPBUF |
| 1033 | 1030 | 6620 | 5F         |        |      | CLRB |         |
| 1034 | 1031 | 6621 | BD         | FBB8   | CLOP | JSR  | INCHA   |
| 1035 | 1032 | 6624 | 81         | 0D     |      | CMPA | #\$0D   |
| 1036 | 1033 | 6626 | 27         | 07     |      | BEQ  | INEND   |
| 1037 | 1034 | 6628 | A7         | 85     |      | STA  | B,X     |
| 1038 | 1035 | 662A | 5C         |        |      | INCB |         |
| 1039 | 1036 | 662B | C1         | 08     |      | CMPB | #8      |
| 1040 | 1037 | 662D | 25         | F2     |      | BLO  | CLOP    |
| 1041 | 1038 |      |            |        |      |      |         |
| 1042 | 1039 | 662F |            | INEND  |      |      |         |
| 1043 | 1040 |      | *shift     |        |      |      |         |
| 1044 | 1041 | 662F | 5D         |        |      | TSTB |         |
| 1045 | 1042 | 6630 | 27         | 11     |      | BEQ  | CL1     |
| 1046 | 1043 | 6632 | 8E         | 6956   |      | LDX  | #INPBUF |

|      |      |      |                 |      |       |       |           |
|------|------|------|-----------------|------|-------|-------|-----------|
| 1047 | 1044 | 6635 | 3A              |      | ABX   |       |           |
| 1048 | 1045 | 6636 | 108E            | 695E |       | LDY   | #INPBUF+8 |
| 1049 | 1046 | 663A | 34              | 04   |       | PSHS  | B         |
| 1050 | 1047 | 663C | A6              | 82   | SFL0P | LDA   | , -X      |
| 1051 | 1048 | 663E | A7              | A2   |       | STA   | , -Y      |
| 1052 | 1049 | 6640 | 5A              |      |       | DECB  |           |
| 1053 | 1050 | 6641 | 26              | F9   |       | BNE   | SFL0P     |
| 1054 | 1051 |      | *clr            |      |       |       |           |
| 1055 | 1052 | 6643 | C6              | 08   | CL1   | LDB   | #8        |
| 1056 | 1053 | 6645 | E0              | E0   |       | SUBB  | , S+      |
| 1057 | 1054 | 6647 | 27              | 0A   |       | BEQ   | CLEND     |
| 1058 | 1055 | 6649 | 8E              | 6956 |       | LDX   | #INPBUF   |
| 1059 | 1056 | 664C | 86              | 30   |       | LDA   | #'0       |
| 1060 | 1057 | 664E | A7              | 80   | CLL0P | STA   | , X+      |
| 1061 | 1058 | 6650 | 5A              |      |       | DECB  |           |
| 1062 | 1059 | 6651 | 26              | FB   |       | BNE   | CLL0P     |
| 1063 | 1060 | 6653 |                 |      |       | CLEND |           |
| 1064 | 1061 |      |                 |      |       |       |           |
| 1065 | 1062 |      | *convert to bin |      |       |       |           |
| 1066 | 1063 | 6653 | C6              | 04   |       | LDB   | #4        |
| 1067 | 1064 | 6655 | 8E              | 6956 |       | LDX   | #INPBUF   |
| 1068 | 1065 | 6658 | 108E            | 6003 |       | LDY   | #SCTN0    |
| 1069 | 1066 | 665C |                 |      | CVT1  |       |           |
| 1070 | 1067 | 665C | A6              | 80   |       | LDA   | , X+      |
| 1071 | 1068 | 665E | 8D              | 1E   |       | BSR   | CVBIN     |
| 1072 | 1069 | 6660 | 25              | 16   |       | BCS   | EREND1    |
| 1073 | 1070 | 6662 | 48              |      |       | LSLA  |           |
| 1074 | 1071 | 6663 | 48              |      |       | LSLA  |           |
| 1075 | 1072 | 6664 | 48              |      |       | LSLA  |           |
| 1076 | 1073 | 6665 | 48              |      |       | LSLA  |           |
| 1077 | 1074 | 6666 | 34              | 02   |       | PSHS  | A         |
| 1078 | 1075 | 6668 | A6              | 80   |       | LDA   | , X+      |
| 1079 | 1076 | 666A | 8D              | 12   |       | BSR   | CVBIN     |
| 1080 | 1077 | 666C | 25              | 08   |       | BCS   | EREND0    |
| 1081 | 1078 | 666E | AB              | E0   |       | ADDA  | , S+      |
| 1082 | 1079 | 6670 | A7              | A0   |       | STA   | , Y+      |
| 1083 | 1080 | 6672 | 5A              |      |       | DECB  |           |
| 1084 | 1081 | 6673 | 26              | E7   |       | BNE   | CVT1      |
| 1085 | 1082 | 6675 | 39              |      |       | RTS   |           |

|      |      |      |    |      |                           |       |         |         |
|------|------|------|----|------|---------------------------|-------|---------|---------|
| 1086 | 1083 |      |    |      |                           |       |         |         |
| 1087 | 1084 | 6676 | 35 | 02   | EREND0                    | PULS  | A       |         |
| 1088 | 1085 | 6678 | 8E | 68F6 | EREND1                    | LDX   | #MSERR  |         |
| 1089 | 1086 | 667B | 8D | 4A   |                           | BSR   | PSTRNG  |         |
| 1090 | 1087 | 667D | 39 |      | EEND                      | RTS   |         |         |
| 1091 | 1088 |      |    |      |                           |       |         |         |
| 1092 | 1089 |      |    |      | *-----                    |       |         |         |
| 1093 | 1090 |      |    |      | *ascii -- bin             |       |         |         |
| 1094 | 1091 | 667E |    |      | CVBIN                     | EQU   | *       |         |
| 1095 | 1092 | 667E | 80 | 30   |                           | SUBA  | #'0     |         |
| 1096 | 1093 | 6680 | 2B | 11   |                           | BMI   | CVERR   |         |
| 1097 | 1094 | 6682 | 81 | 09   |                           | CMPA  | #9      |         |
| 1098 | 1095 | 6684 | 23 | 0A   |                           | BLS   | CVBEND  |         |
| 1099 | 1096 | 6686 | 81 | 11   |                           | CMPA  | #\$11   |         |
| 1100 | 1097 | 6688 | 25 | 09   |                           | BLO   | CVERR   |         |
| 1101 | 1098 | 668A | 81 | 16   |                           | CMPA  | #\$16   |         |
| 1102 | 1099 | 668C | 22 | 05   |                           | BHI   | CVERR   |         |
| 1103 | 1100 | 668E | 80 | 07   |                           | SUBA  | #7      |         |
| 1104 | 1101 | 6690 |    |      | CVBEND                    |       |         |         |
| 1105 | 1102 | 6690 | 1C | FE   |                           | ANDCC | #\$FE   | reset C |
| 1106 | 1103 | 6692 | 39 |      |                           | RTS   |         |         |
| 1107 | 1104 | 6693 | 1A | 01   | CVERR                     | ORCC  | #\$01   | set C   |
| 1108 | 1105 | 6695 | 39 |      |                           | RTS   |         |         |
| 1109 | 1106 |      |    |      |                           |       |         |         |
| 1110 | 1107 |      |    |      | *-----                    |       |         |         |
| 1111 | 1108 |      |    |      | *add 2bytes to 4bytes     |       |         |         |
| 1112 | 1109 |      |    |      | * X,X+1,X+2,X+3 + Y,Y+1   |       |         |         |
| 1113 | 1110 |      |    |      | * data X is big endian    |       |         |         |
| 1114 | 1111 |      |    |      | * data Y is little endian |       |         |         |
| 1115 | 1112 | 6696 |    |      | ADD42                     | EQU   | *       |         |
| 1116 | 1113 | 6696 | 34 | 36   |                           | PSHS  | A,B,X,Y |         |
| 1117 | 1114 | 6698 | 32 | 7C   |                           | LEAS  | -4,S    |         |
| 1118 | 1115 | 669A | C6 | 04   |                           | LDB   | #4      |         |
| 1119 | 1116 | 669C | A6 | 80   | AD1                       | LDA   | ,X+     |         |
| 1120 | 1117 | 669E | A7 | E0   |                           | STA   | ,S+     |         |
| 1121 | 1118 | 66A0 | 5A |      |                           | DECB  |         |         |
| 1122 | 1119 | 66A1 | 26 | F9   |                           | BNE   | AD1     |         |
| 1123 | 1120 |      |    |      |                           |       |         |         |
| 1124 | 1121 |      |    |      | *add (lower)              |       |         |         |

|      |      |      |                        |        |       |      |         |
|------|------|------|------------------------|--------|-------|------|---------|
| 1125 | 1122 | 66A3 | A6                     | A0     |       | LDA  | ,Y+     |
| 1126 | 1123 | 66A5 | AB                     | E2     |       | ADDA | , -S    |
| 1127 | 1124 | 66A7 | A7                     | E4     |       | STA  | ,S      |
| 1128 | 1125 | 66A9 | 24                     | 02     |       | BCC  | AD2     |
| 1129 | 1126 | 66AB | 6C                     | 7F     |       | INC  | -1,S    |
| 1130 | 1127 |      | *add (upper)           |        |       |      |         |
| 1131 | 1128 | 66AD | A6                     | A4     | AD2   | LDA  | ,Y      |
| 1132 | 1129 | 66AF | AB                     | E2     |       | ADDA | , -S    |
| 1133 | 1130 | 66B1 | A7                     | E4     |       | STA  | ,S      |
| 1134 | 1131 | 66B3 | 24                     | 02     |       | BCC  | AD3     |
| 1135 | 1132 | 66B5 | 6C                     | 7F     |       | INC  | -1,S    |
| 1136 | 1133 | 66B7 | 30                     | 1C     | AD3   | LEAX | -4,X    |
| 1137 | 1134 | 66B9 | 32                     | 7E     |       | LEAS | -2,S    |
| 1138 | 1135 | 66BB | C6                     | 04     |       | LDB  | #4      |
| 1139 | 1136 | 66BD | A6                     | E0     | AD4   | LDA  | ,S+     |
| 1140 | 1137 | 66BF | A7                     | 80     |       | STA  | ,X+     |
| 1141 | 1138 | 66C1 | 5A                     |        |       | DECB |         |
| 1142 | 1139 | 66C2 | 26                     | F9     |       | BNE  | AD4     |
| 1143 | 1140 | 66C4 | 35                     | 36     |       | PULS | A,B,X,Y |
| 1144 | 1141 | 66C6 | 39                     |        |       | RTS  |         |
| 1145 | 1142 |      |                        |        |       |      |         |
| 1146 | 1143 |      | *-----                 |        |       |      |         |
| 1147 | 1144 |      | * pstring              |        |       |      |         |
| 1148 | 1145 | 66C7 |                        | PSTRNG | EQU   | *    |         |
| 1149 | 1146 | 66C7 | 34                     | 02     |       | PSHS | A       |
| 1150 | 1147 | 66C9 | A6                     | 80     | PSLOP | LDA  | ,X+     |
| 1151 | 1148 | 66CB | 81                     | 04     |       | CMPA | #4      |
| 1152 | 1149 | 66CD | 27                     | 05     |       | BEQ  | PSEND   |
| 1153 | 1150 | 66CF | BD                     | FBBB   |       | JSR  | OUTCHA  |
| 1154 | 1151 | 66D2 | 20                     | F5     |       | BRA  | PSLOP   |
| 1155 | 1152 | 66D4 | 35                     | 02     | PSEND | PULS | A       |
| 1156 | 1153 | 66D6 | 39                     |        |       | RTS  |         |
| 1157 | 1154 |      |                        |        |       |      |         |
| 1158 | 1155 |      | *-----                 |        |       |      |         |
| 1159 | 1156 |      | *disp MM-DD-YY         |        |       |      |         |
| 1160 | 1157 |      | *indata A(Low),B(High) |        |       |      |         |
| 1161 | 1158 | 66D7 |                        | DSPYMD | EQU   | *    |         |
| 1162 | 1159 | 66D7 | 1E                     | 89     |       | EXG  | A,B     |
| 1163 | 1160 | 66D9 | 34                     | 10     |       | PSHS | X       |

|      |      |      |    |                |           |        |             |  |
|------|------|------|----|----------------|-----------|--------|-------------|--|
| 1164 | 1161 | 66DB | 34 | 06             | PSHS      | A,B    |             |  |
| 1165 | 1162 | 66DD | C4 | E0             | ANDB      | #\$E0  |             |  |
| 1166 | 1163 | 66DF | 54 |                | LSRB      |        |             |  |
| 1167 | 1164 | 66E0 | 54 |                | LSRB      |        |             |  |
| 1168 | 1165 | 66E1 | 54 |                | LSRB      |        |             |  |
| 1169 | 1166 | 66E2 | 54 |                | LSRB      |        |             |  |
| 1170 | 1167 | 66E3 | 54 |                | LSRB      |        |             |  |
| 1171 | 1168 | 66E4 | 47 |                | ASRA      |        |             |  |
| 1172 | 1169 | 66E5 | 24 | 02             | BCC       | DYMD1  |             |  |
| 1173 | 1170 | 66E7 | CB | 08             | ADDB      | #8     |             |  |
| 1174 | 1171 | 66E9 | F7 | 601A           | DYMD1 STB | WMDY   | MM          |  |
| 1175 | 1172 | 66EC | 35 | 06             | PULS      | A,B    |             |  |
| 1176 | 1173 | 66EE | 34 | 04             | PSHS      | B      |             |  |
| 1177 | 1174 | 66F0 | C4 | 1F             | ANDB      | #\$1F  |             |  |
| 1178 | 1175 | 66F2 | F7 | 601B           | STB       | WMDY+1 | DD          |  |
| 1179 | 1176 | 66F5 | 35 | 04             | PULS      | B      |             |  |
| 1180 | 1177 |      |    |                |           |        |             |  |
| 1181 | 1178 | 66F7 | 44 |                | LSRA      |        |             |  |
| 1182 | 1179 |      | *  | ASLB           |           |        |             |  |
| 1183 | 1180 |      | *  | BCC DYMD2      |           |        |             |  |
| 1184 | 1181 |      | *  | INCA           |           |        |             |  |
| 1185 | 1182 | 66F8 |    | DYMD2          |           |        |             |  |
| 1186 | 1183 | 66F8 | 80 | 14             | SUBA      | #20    | A+1980-2000 |  |
| 1187 | 1184 | 66FA | B7 | 601C           | STA       | WMDY+2 | YY          |  |
| 1188 | 1185 |      |    |                |           |        |             |  |
| 1189 | 1186 |      |    | *disp MM-DD-YY |           |        |             |  |
| 1190 | 1187 | 66FD | 86 | 20             | LDA       | #'     |             |  |
| 1191 | 1188 | 66FF | BD | FBBB           | JSR       | OUTCHA |             |  |
| 1192 | 1189 | 6702 | 8E | 601A           | LDX       | #WMDY  |             |  |
| 1193 | 1190 | 6705 | A6 | 80             | LDA       | ,X+    |             |  |
| 1194 | 1191 | 6707 | 8D | 14             | BSR       | DB2ASC | MM          |  |
| 1195 | 1192 | 6709 | 86 | 2D             | LDA       | #' -   |             |  |
| 1196 | 1193 | 670B | BD | FBBB           | JSR       | OUTCHA |             |  |
| 1197 | 1194 | 670E | A6 | 80             | LDA       | ,X+    |             |  |
| 1198 | 1195 | 6710 | 8D | 0B             | BSR       | DB2ASC | DD          |  |
| 1199 | 1196 | 6712 | 86 | 2D             | LDA       | #' -   |             |  |
| 1200 | 1197 | 6714 | BD | FBBB           | JSR       | OUTCHA |             |  |
| 1201 | 1198 | 6717 | A6 | 80             | LDA       | ,X+    |             |  |
| 1202 | 1199 | 6719 | 8D | 02             | BSR       | DB2ASC | YY          |  |

|      |      |      |                  |        |        |      |         |              |
|------|------|------|------------------|--------|--------|------|---------|--------------|
| 1203 | 1200 | 671B | 35               | 90     |        | PULS | X,PC    |              |
| 1204 | 1201 |      |                  |        |        |      |         |              |
| 1205 | 1202 |      | *                |        |        |      |         |              |
| 1206 | 1203 |      | *-----           |        |        |      |         |              |
| 1207 | 1204 |      | *(A:BIN) --ASCII |        |        |      |         |              |
| 1208 | 1205 | 671D |                  | DB2ASC | EQU    | *    |         |              |
| 1209 | 1206 | 671D | 34               | 06     |        | PSHS | A,B     |              |
| 1210 | 1207 | 671F | 1F               | 89     |        | TFR  | A,B     |              |
| 1211 | 1208 | 6721 | 86               | 30     |        | LDA  | #\$30   |              |
| 1212 | 1209 | 6723 | C1               | 0A     | DBLOP  | CMPB | #10     |              |
| 1213 | 1210 | 6725 | 25               | 05     |        | BLO  | DB2     |              |
| 1214 | 1211 | 6727 | C0               | 0A     |        | SUBB | #10     |              |
| 1215 | 1212 | 6729 | 4C               |        |        | INCA |         |              |
| 1216 | 1213 | 672A | 20               | F7     |        | BRA  | DBLOP   |              |
| 1217 | 1214 | 672C | BD               | FBBB   | DB2    | JSR  | OUTCHA  |              |
| 1218 | 1215 | 672F | 1F               | 98     |        | TFR  | B,A     |              |
| 1219 | 1216 | 6731 | 8B               | 30     |        | ADDA | #\$30   |              |
| 1220 | 1217 | 6733 | BD               | FBBB   |        | JSR  | OUTCHA  |              |
| 1221 | 1218 | 6736 | 35               | 86     |        | PULS | A,B,PC  |              |
| 1222 | 1219 |      |                  |        |        |      |         |              |
| 1223 | 1220 |      | *disp            | FILE   | size   |      |         |              |
| 1224 | 1221 | 6738 |                  | DSPSIZ | EQU    | *    |         |              |
| 1225 | 1222 | 6738 | 34               | 10     |        | PSHS | X       |              |
| 1226 | 1223 | 673A | BE               | 601D   |        | LDX  | WFSIZ   |              |
| 1227 | 1224 | 673D | 27               | 12     |        | BEQ  | DSS1    | if 64k over? |
| 1228 | 1225 | 673F | 86               | 2D     |        | LDA  | #' -    |              |
| 1229 | 1226 | 6741 | BD               | FBBB   |        | JSR  | OUTCHA  |              |
| 1230 | 1227 | 6744 | BD               | FBBB   |        | JSR  | OUTCHA  |              |
| 1231 | 1228 | 6747 | BD               | FBBB   |        | JSR  | OUTCHA  |              |
| 1232 | 1229 | 674A | 86               | 6B     |        | LDA  | #' k    |              |
| 1233 | 1230 | 674C | BD               | FBBB   |        | JSR  | OUTCHA  |              |
| 1234 | 1231 | 674F | 20               | 3B     |        | BRA  | DSEND   |              |
| 1235 | 1232 | 6751 |                  | DSS1   |        |      |         |              |
| 1236 | 1233 | 6751 | B6               | 601F   |        | LDA  | WFSIZ+2 |              |
| 1237 | 1234 | 6754 | 81               | 64     |        | CMPA | #100    |              |
| 1238 | 1235 | 6756 | 23               | 12     |        | BLS  | DSS13   |              |
| 1239 | 1236 | 6758 | 1F               | 89     |        | TFR  | A,B     |              |
| 1240 | 1237 | 675A | 86               | 30     |        | LDA  | #\$30   |              |
| 1241 | 1238 | 675C | C1               | 64     | DSL0P1 | CMPB | #100    |              |

|      |      |      |    |        |        |      |        |
|------|------|------|----|--------|--------|------|--------|
| 1242 | 1239 | 675E | 25 | 05     |        | BLO  | DSS12  |
| 1243 | 1240 | 6760 | C0 | 64     |        | SUBB | #100   |
| 1244 | 1241 | 6762 | 4C |        |        | INCA |        |
| 1245 | 1242 | 6763 | 20 | F7     |        | BRA  | DSL0P1 |
| 1246 | 1243 | 6765 | BD | FBBB   | DSS12  | JSR  | OUTCHA |
| 1247 | 1244 | 6768 | 20 | 05     |        | BRA  | DSS14  |
| 1248 | 1245 |      |    |        |        |      |        |
| 1249 | 1246 | 676A | 86 | 20     | DSS13  | LDA  | #'     |
| 1250 | 1247 | 676C | BD | FBBB   |        | JSR  | OUTCHA |
| 1251 | 1248 | 676F | 86 | 30     | DSS14  | LDA  | #\$30  |
| 1252 | 1249 | 6771 | C1 | 0A     | DSL0P2 | CMPB | #10    |
| 1253 | 1250 | 6773 | 25 | 05     |        | BLO  | DSS2   |
| 1254 | 1251 | 6775 | C0 | 0A     |        | SUBB | #10    |
| 1255 | 1252 | 6777 | 4C |        |        | INCA |        |
| 1256 | 1253 | 6778 | 20 | F7     |        | BRA  | DSL0P2 |
| 1257 | 1254 | 677A | 81 | 30     | DSS2   | CMPA | #\$30  |
| 1258 | 1255 | 677C | 26 | 01     |        | BNE  | DSS3   |
| 1259 | 1256 | 677E | 4C |        |        | INCA |        |
| 1260 | 1257 | 677F | BD | FBBB   | DSS3   | JSR  | OUTCHA |
| 1261 | 1258 | 6782 | 86 | 6B     |        | LDA  | #'k    |
| 1262 | 1259 | 6784 | BD | FBBB   |        | JSR  | OUTCHA |
| 1263 | 1260 | 6787 | 86 | 20     |        | LDA  | #'     |
| 1264 | 1261 | 6789 | BD | FBBB   |        | JSR  | OUTCHA |
| 1265 | 1262 | 678C | 35 | 90     | DSEND  | PULS | X,PC   |
| 1266 | 1263 |      |    |        |        |      |        |
| 1267 | 1264 |      |    | *----- |        |      |        |
| 1268 | 1265 | 678E |    | DSPCLS | EQU    | *    |        |
| 1269 | 1266 | 678E | 34 | 10     |        | PSHS | X      |
| 1270 | 1267 | 6790 | 86 | 20     |        | LDA  | #'     |
| 1271 | 1268 | 6792 | BD | FBBB   |        | JSR  | OUTCHA |
| 1272 | 1269 | 6795 | 8E | 6021   |        | LDX  | #CLSTH |
| 1273 | 1270 | 6798 | C6 | 02     |        | LDB  | #2     |
| 1274 | 1271 | 679A | A6 | 80     | DCL0P1 | LDA  | ,X+    |
| 1275 | 1272 | 679C | 17 | FF7E   |        | LBSR | DB2ASC |
| 1276 | 1273 | 679F | 5A |        |        | DECB |        |
| 1277 | 1274 | 67A0 | 26 | F8     |        | BNE  | DCL0P1 |
| 1278 | 1275 | 67A2 | 86 | 20     |        | LDA  | #'     |
| 1279 | 1276 | 67A4 | BD | FBBB   |        | JSR  | OUTCHA |
| 1280 | 1277 | 67A7 | C6 | 02     |        | LDB  | #2     |

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1281 1278 67A9 A6 80 DCL0P2 LDA ,X+
1282 1279 67AB 17 FF6F LBSR DB2ASC
1283 1280 67AE 5A DECB
1284 1281 67AF 26 F8 BNE DCL0P2
1285 1282 67B1 35 90 PULS X,PC
1286 1283
1287 1284 *****
1288 1285 *===== parameter =====
1289 1286 67B3 40 CMD0 FCB $40,$00,$00,$00,$00,$95 '@
1290 1287 67B9 41 CMD1 FCB $41,$00,$00,$00,$00,$FF 'A
1291 1288 67BF 51 CMD17 FCB $51,$00,$00,$00,$00,$FF 'Q
1292 1289 67C5 58 CMD24 FCB $58,$00,$00,$00,$00,$FF 'X
1293 1290 67CB 48 CMD8 FCB $48,$00,$00,$01,$AA,$87 'H
1294 1291 67D1 7A CMD58 FCB $7A,$00,$00,$00,$00,$FF 'z
1295 1292 67D7 77 CMD55 FCB $77,$00,$00,$00,$00,$FF 'w
1296 1293 67DD 69 ACMD41 FCB $69,$40,$FF,$80,$00,$FF 'i
1297 1294 67E3 49 CMD9 FCB $49,$00,$00,$00,$00,$FF 'I
1298 1295 *
1299 1296 67E9 43 MCMD0 FCC /CMD0/, $D, $A
1300 1297 67EF 04 FCB 4
1301 1298 67F0 43 MCMD8 FCC /CMD8/, $D, $A
1302 1299 67F6 04 FCB 4
1303 1300 67F7 43 MCMD58 FCC /CMD58/, $D, $A
1304 1301 67FE 04 FCB 4
1305 1302 67FF 43 MCMD55 FCC /CMD55/, $D, $A
1306 1303 6806 04 FCB 4
1307 1304 6807 41 MCMD41 FCC /ACMD41/, $D, $A
1308 1305 680F 04 FCB 4
1309 1306 6810 43 MCMD9 FCC /CMD9/, $D, $A
1310 1307 6816 04 FCB 4
1311 1308 *
1312 1309 6817 0D MSMENU1 FCC $0D,$0A,/1:Read Sct, 2:Write
1313 Sct, 3:Get SD data, 4:SD Init, 5:Disp MBR, /
1314 1310 6858 04 FCB 4
1315 1311 6859 0D MSMENU2 FCC $0D,$0A,/6:Disp BPB, 7:Disp
1316 FAT, 8:Disp DIR, 9:Load FILE, 0:End : /
1317 1312 6894 04 FCB 4
1318 1313 6895 53 MSSCT FCC /SctNo: /
1319 1314 689C 04 FCB 4

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1320      1315  689D  57      MSWROK  FCC      /Write OK?: /
1321      1316  68A8  04              FCB      4
1322      1317  68A9  0D      MSSD0   FCC      $0D,$0A,/SD Card/,$0D,$0A
1323      1318  68B4  04              FCB      4
1324      1319  68B5  0D      MSSD1   FCC      $0D,$0A,/SDHC Card/,$0D,$0A
1325      1320  68C2  04              FCB      4
1326      1321  68C3  0D      MSINIT  FCC      $0D,$0A,$0D,$0A,/SD Card
1327      initializing.../
1328      1322  68DE  04              FCB      4
1329      1323  68DF  53      MERR     FCC      /SD Initialize
1330      error!/, $0D,$0A
1331      1324  68F5  04              FCB      4
1332      1325  68F6  49      MSERR    FCC      /Input error!/, $0D,$0A
1333      1326  6904  04              FCB      4
1334      1327  6905  0D      MSOFS    FCC      $0D,$0A,/Sector no
1335      is:/, $0D,$0A
1336      1328  6916  04              FCB      4
1337      1329  6917  0D      MSBPB    FCC      $0D,$0A,/ BPB: /
1338      1330  691F  04              FCB      4
1339      1331  6920  0D      MSFAT    FCC      $0D,$0A,/ FAT: /
1340      1332  6928  04              FCB      4
1341      1333  6929  0D      MSDIR    FCC      $0D,$0A,/ DIR: /
1342      1334  6931  04              FCB      4
1343      1335  6932  0D      MSCL2    FCC      $0D,$0A,/ 1stCLUSTER: /
1344      1336  6941  04              FCB      4
1345      1337  6942  0D      MSINFL   FCC      $0D,$0A,/Select File No : /
1346      1338  6955  04              FCB      4
1347      1339
1348      1340      *****
1349      1341  6956  FF      INPBUF   RMB      8
1350      1342
1351      1343      END      START
1352      0 error(s), 0 warning(s) is detected.
1353      Label address
1354      ACMD41  67DD  AD1      669C  AD2      66AD  AD3      66B7  AD4
1355      66BD
1356      ADD42   6696  ASC411  6169  ASCMD41  615C  BI21      65FD  BI22
1357      6603
1358

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|      |        |      |        |      |        |      |        |      |        |
|------|--------|------|--------|------|--------|------|--------|------|--------|
| 1359 | BUSY   | 64FF | CALCH1 | 65CF | CALCHC | 65C6 | CALCSD | 65BC | CKRS   |
| 1360 | 64F2   |      |        |      |        |      |        |      |        |
| 1361 | CL1    | 6643 | CLEND  | 6653 | CLLOP  | 664E | CLOP   | 6621 | CLSTH  |
| 1362 | 6021   |      |        |      |        |      |        |      |        |
| 1363 | CLSTL  | 6023 | CMD0   | 67B3 | CMD1   | 67B9 | CMD17  | 67BF | CMD24  |
| 1364 | 67C5   |      |        |      |        |      |        |      |        |
| 1365 | CMD55  | 67D7 | CMD58  | 67D1 | CMD8   | 67CB | CMD9   | 67E3 | CPY1   |
| 1366 | 6516   |      |        |      |        |      |        |      |        |
| 1367 | CPY2   | 64A9 | CRLF   | 6612 | CVBEND | 6690 | CVBIN  | 667E | CVERR  |
| 1368 | 6693   |      |        |      |        |      |        |      |        |
| 1369 | CVT1   | 665C | DB2    | 672C | DB2ASC | 671D | DBLOP  | 6723 | DCL0P1 |
| 1370 | 679A   |      |        |      |        |      |        |      |        |
| 1371 | DCL0P2 | 67A9 | DSEND  | 678C | DSL0P1 | 675C | DSL0P2 | 6771 | DSPB1  |
| 1372 | 6552   |      |        |      |        |      |        |      |        |
| 1373 | DSPB2  | 6554 | DSPB3  | 6565 | DSPBIN | 65E9 | DSPBPB | 634A | DSPBUF |
| 1374 | 6548   |      |        |      |        |      |        |      |        |
| 1375 | DSPC1  | 63D6 | DSPCLS | 678E | DSPDIR | 636E | DSPEND | 63F3 | DSPFAT |
| 1376 | 635C   |      |        |      |        |      |        |      |        |
| 1377 | DSPFL1 | 6382 | DSPFL2 | 638F | DSPFL3 | 639A | DSPFL4 | 63A9 | DSPMBR |
| 1378 | 62F0   |      |        |      |        |      |        |      |        |
| 1379 | DSPSIZ | 6738 | DSPYMD | 66D7 | DSS1   | 6751 | DSS12  | 6765 | DSS13  |
| 1380 | 676A   |      |        |      |        |      |        |      |        |
| 1381 | DSS14  | 676F | DSS2   | 677A | DSS3   | 677F | DYMD1  | 66E9 | DYMD2  |
| 1382 | 66F8   |      |        |      |        |      |        |      |        |
| 1383 | EEND   | 667D | EREND0 | 6676 | EREND1 | 6678 | GET1   | 62FB | GET2   |
| 1384 | 62FD   |      |        |      |        |      |        |      |        |
| 1385 | GET8CH | 661D | GETBPB | 6331 | GETD1  | 6586 | GETDAT | 6582 | GETMBR |
| 1386 | 630E   |      |        |      |        |      |        |      |        |
| 1387 | GETSD  | 624F | GPL0P  | 6314 | GPL0P2 | 633C | GR00   | 60D4 | GR2580 |
| 1388 | 6196   |      |        |      |        |      |        |      |        |
| 1389 | GR2581 | 61AD | GR2582 | 61AF | GR410  | 6171 | GR550  | 6152 | GR580  |
| 1390 | 6128   |      |        |      |        |      |        |      |        |
|      | GR581  | 6132 | GR582  | 6134 | GR80   | 60F8 | GR81   | 610A | GR90   |
|      | 6262   |      |        |      |        |      |        |      |        |
|      | GR91   | 6271 | IL0P1  | 61D2 | IL0P2  | 61E5 | IL0P3  | 61FF | IL0P4  |
|      | 622D   |      |        |      |        |      |        |      |        |
|      | INCHA  | FBB8 | INEND  | 662F | INIT   | 604D | INPBUF | 6956 | INSCT  |
|      | 659C   |      |        |      |        |      |        |      |        |

|        |      |        |      |         |      |         |      |         |
|--------|------|--------|------|---------|------|---------|------|---------|
| ISTT   | 60BA | LD1    | 6412 | LD2     | 641A | LD3     | 6432 | LD4     |
| 643D   |      |        |      |         |      |         |      |         |
| LD5    | 6449 | LD51   | 6457 | LD52    | 645F | LD53    | 6467 | LD6     |
| 6473   |      |        |      |         |      |         |      |         |
| LDFILE | 63F4 | LDL0P  | 6407 | MAIN    | 6029 | MCMD0   | 67E9 | MCMD41  |
| 6807   |      |        |      |         |      |         |      |         |
| MCMD55 | 67FF | MCMD58 | 67F7 | MCMD8   | 67F0 | MCMD9   | 6810 | MEN1    |
| 6066   |      |        |      |         |      |         |      |         |
| MEN2   | 606E | MEN3   | 6076 | MEN4    | 607E | MEN5    | 6085 | MEN6    |
| 608D   |      |        |      |         |      |         |      |         |
| MEN7   | 6095 | MEN8   | 609D | MEN9    | 60A5 | MENU    | 604F | MERR    |
| 68DF   |      |        |      |         |      |         |      |         |
| MSBPB  | 6917 | MSCL2  | 6932 | MSDIR   | 6929 | MSERR   | 68F6 | MSFAT   |
| 6920   |      |        |      |         |      |         |      |         |
| MSINFL | 6942 | MSINIT | 68C3 | MSMENU1 | 6817 | MSMENU2 | 6859 | MS0FS   |
| 6905   |      |        |      |         |      |         |      |         |
| MSSCT  | 6895 | MSSD0  | 68A9 | MSSD1   | 68B5 | MSWROK  | 689D | OKEND   |
| 62DC   |      |        |      |         |      |         |      |         |
| OUTCHA | FBBB | PBCTL  | FD9F | PBDAT   | FD9E | PSEND   | 66D4 | PSL0P   |
| 66C9   |      |        |      |         |      |         |      |         |
| PSTRNG | 66C7 | RD1    | 6525 | RD1SCT  | 650D | RD2     | 652C | RD3     |
| 653C   |      |        |      |         |      |         |      |         |
| RESDAT | 6007 | RPT41  | 613D | RWBUF   | 6E00 | SC01    | 60CB | SC2581  |
| 618E   |      |        |      |         |      |         |      |         |
| SC551  | 614A | SC581  | 6120 | SC80    | 60F0 | SC91    | 625A | SCL0P1  |
| 6291   |      |        |      |         |      |         |      |         |
| SCL0P2 | 62A4 | SCL0P3 | 62B7 | SCL0P4  | 62CA | SCMD0   | 60C3 | SCMD258 |
| 6181   |      |        |      |         |      |         |      |         |
| SCMD55 | 613D | SCMD58 | 6113 | SCMD8   | 60E3 | SCTBPB  | 6009 | SCTCLT  |
| 6015   |      |        |      |         |      |         |      |         |
| SCTDIR | 6011 | SCTFAT | 600D | SCTN0   | 6003 | SCTN1   | 6004 | SCTN2   |
| 6005   |      |        |      |         |      |         |      |         |
| SCTN3  | 6006 | SCTPCL | 6019 | SD1     | 61C3 | SDERR   | 62D6 | SDHC    |
| 6008   |      |        |      |         |      |         |      |         |
| SDINIT | 60AD | SDRD   | 62DD | SDW1    | 64B8 | SDW2    | 64C3 | SDWR1   |
| 649A   |      |        |      |         |      |         |      |         |
| SDWRT  | 648A | SELOP  | 6326 | SETCKH  | 65E0 | SETCKL  | 65D7 | SETSCT  |
| 65A4   |      |        |      |         |      |         |      |         |

|        |      |        |      |        |      |       |      |        |
|--------|------|--------|------|--------|------|-------|------|--------|
| SETSN0 | 6320 | SFL0P  | 663C | SFT1   | 65B2 | SKP1  | 6211 | SKP2   |
| 6246   |      |        |      |        |      |       |      |        |
| SNDD1  | 656D | SNDDAT | 6566 | SNDDT1 | 64E4 | START | 6000 | STTADR |
| 6000   |      |        |      |        |      |       |      |        |
| VN     | 6002 | WFSIZ  | 601D | WK1    | 6025 | WK2   | 6027 | WMDY   |
| 601A   |      |        |      |        |      |       |      |        |

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SDHC8FM

Since the list is long, I've added vertical scrolling this time. There's no "Select All" button, but you can double-click items in the list and copy them with Ctrl-C.